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Design and Evaluation of a Smart Scheduling Function for Swedish Preschools

A study of design and implementation for an algorithm-based scheduling solution

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Abstract

This thesis examines how algorithmic decision support can be integrated into the scheduling practices of Swedish preschools. Scheduling is a sociotechnical activity shaped by legal requirements, pedagogical rhythms, fairness norms, and tacit professional knowledge. Through three iterative cycles of design, prototyping, and participatory evaluation, the study explores how these constraints can be formalised computationally and how algorithmic proposals can be made transparent and configurable for users of *Tempus Admin*.

A rule-based prototype was developed using hard and soft constraints with adjustable weights. The system explains decisions, visualises rule conflicts, and allows users to compare multiple scheduling alternatives. Practitioner evaluations show that algorithmic scheduling is most valuable when it supports professional judgement, reduces administrative work, and makes reasoning visible, rather than when it maximises optimisation.

The study finds that fairness is understood relationally, that weekly stability is essential, and that hybrid scheduling strategies are preferred. The thesis contributes: a transparent constraint model, empirical insights on explainability and user control, and design principles for sociotechnical scheduling systems.

Keywords: preschool scheduling, sociotechnical systems, algorithmic decision support, constraint modelling, explainability, participatory design, human–computer interaction

Sammanfattning

Denna avhandling undersöker hur algoritmiskt beslutsstöd kan integreras i förskolans schemaläggning. Schemaläggning är en socioteknisk praktik där juridiska krav, pedagogiska rytmer, rättvisenormer och tyst yrkeskunskap samverkar. Genom tre iterativa cykler av design, prototypande och deltagande utvärdering studeras hur dessa villkor kan formaliseras i en beräkningsmodell och hur algoritmiska förslag kan göras begripliga och konfigurerbara för användare av *Tempus Admin*.

Arbetet resulterade i en regelbaserad prototyp med hårda och mjuka begränsningar och justerbara vikter. Systemet förklarar beslut, visualiserar regelkonflikter och möjliggör jämförelse mellan flera schemalternativ. Utvärderingarna visar att algoritmiskt stöd är mest värdefullt när det stärker professionellt omdöme, minskar administration och gör schemaläggningens logik synlig snarare än när det enbart optimerar.

Studien visar att rättvisa uppfattas relationellt, att veckovis stabilitet är central och att hybrida strategier föredras. Avhandlingen bidrar med en transparent villkorsmodell, empiriska insikter om förklarbarhet och användarkontroll samt designprinciper för sociotekniska schemaläggningssystem.

Nyckelord: förskola, schemaläggning, sociotekniska system, algoritmiskt beslutsstöd, begränsings-modellering, förklarbarhet, deltagande design, människa–datorinteraktion

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Edvin Kurtsson
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List of Abbreviations

AI Artificial Intelligence

DSR Design Science Research

HCI Human–Computer Interaction

PD Participatory Design

GDPR General Data Protection Regulation

SKR Sveriges Kommuner och Regioner (Swedish Association of Local Authorities and Regions)

EU AI Act European Union Artificial Intelligence Act

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1 Introduction

Efficient staff scheduling is a recurring challenge across many sectors, yet it is often perceived as particularly demanding in preschool education. Swedish preschools must balance multiple, and sometimes competing, requirements: labour law and collective agreements, pedagogical goals such as ensuring complete work teams during core hours, and the individual preferences of staff members. Schedulers frequently rely on spreadsheets, paper templates, or fragmented digital tools to construct recurring schedules. This results in high administrative workload, limited transparency in decision-making, and difficulties in responding quickly to unexpected events such as staff absences or sudden demand spikes.

In preschools, scheduling is not merely an operational activity but one that directly influences the quality of education and care. The presence of qualified preschool teachers during core pedagogical hours is mandated by law, while equitable rotation of early and late shifts affects both staff wellbeing and perceived fairness among colleagues. Poorly balanced schedules may contribute to stress, inconsistent child–teacher relationships, and reduced time for pedagogical planning. At a societal level, efficient and fair scheduling supports the sustainability of Sweden's welfare model, in which accessible, high-quality early childhood education is both a right and a cornerstone of gender equality policy [1, 2].

At the same time, advances in optimisation and so called smart algorithms offer new opportunities to support or automate parts of the scheduling process. Similar approaches have been successfully applied in domains such as healthcare, transportation, and workforce management to reduce scheduling effort and improve the quality of resulting timetables [3, 4]. However, transferring these methods to preschools requires sensitivity to pedagogical and legal frameworks as well as a strong emphasis on transparency, fairness, and user control to maintain trust among practitioners and unions. Preschool scheduling is also distinct in its cultural and ethical dimensions: decisions must balance formal compliance with local norms of fairness between colleagues and a child-centred pedagogy.

Despite widespread digitalisation efforts in the Swedish education sector, most current tools stop short of providing algorithmic decision support. Platforms

such as *Tempus*, *Tieto Education*, and *Vklass* assist with attendance tracking and administrative reporting but still depend on manual input to construct coherent staff plans [2]. This gap illustrates a broader challenge in the public sector. While data availability has improved, intelligent decision support functions remain underdeveloped.

The focus of this work is the design and evaluation of a smart scheduling function to be integrated within *Tempus Admin*. The function aims to automatically generate rolling staff schedules based on predefined rules and historical data, while allowing administrators to review, adjust, and understand the system's reasoning. By combining algorithmic reasoning with explainable interfaces and participatory design, the project seeks to demonstrate how smart automation can assist human schedulers without replacing their professional judgement.

1.1 Towards the Research Aim

Building on the challenges outlined above, this thesis focuses on the iterative design and evaluation of a smart, algorithm-supported scheduling system for Swedish preschools. The purpose of the study is to examine how different scheduling strategies and explanation features shape practitioners' perceptions of fairness, usability, and trust, while supporting pedagogical continuity and professional judgement.

Iteration is treated as a central analytical lens in the study, as the evolution of the system across design cycles provides insight into how algorithmic decision support is interpreted and negotiated in practice.

The study is guided by the following research questions:

- **RQ1:** How do preschool schedulers assess the fairness, predictability, and pedagogical continuity of scheduling outcomes produced throughout the iterative design process?
- **RQ2:** How do explainability features and visible rule feedback influence users' trust, confidence, and accountability when interpreting system-generated schedules?
- **RQ3:** How does iterative participatory prototyping affect the system's

ability to reduce administrative workload while preserving practitioner control and supporting tacit professional judgement?

Through iterative prototyping and qualitative user evaluation, the project contributes to applied HCI research and the broader field of responsible algorithmic design in public administration. By treating transparency, fairness, and configurability as core design principles rather than optional additions, the project demonstrates how human-centred algorithmic systems can strengthen professional decision making in early childhood education.

1.2 External partner

This project is carried out in collaboration with *Tempus*, a Swedish company that develops digital platforms for preschools and after-school care. Tempus provides three main modules: *Tempus Hemma*, used by guardians to register planned attendance; *Tempus Nu*, which enables staff to check children in and out in real time; and *Tempus Admin*, a platform for administrators and schedulers to manage attendance and staffing data. The system is widely adopted across Sweden, serving over 100 municipalities and independent providers, and supports more than 350,000 users [5].

2 Background

Staff scheduling in Swedish preschools operates within a tightly regulated and socially significant domain. Unlike scheduling in industrial or commercial settings, preschool rostering must satisfy pedagogical, legal, and ethical obligations that reflect Sweden's welfare-state model. Preschools are governed by the Education Act (*Skollagen*, SFS 2010:800), which stipulates that every child should have access to a safe and stimulating environment that promotes learning and development. The Curriculum for the Preschool (Lpfö 18) further emphasises pedagogical continuity, teamwork, and equitable opportunities for children throughout the day [1]. This implies not only legal staffing ratios but also the presence of qualified educators during core pedagogical hours, requiring schedules that balance staff availability, competence, and time for pedagogical planning.

In parallel, the Work Environment Act (*Arbetsmiljölagen*) and collective agreements negotiated between Kommunal and Sveriges Kommuner och Regioner (SKR) regulate working time, breaks, rest periods, and the equitable rotation of early and late shifts [6]. These overlapping frameworks create a dense web of constraints that schedulers must navigate. Municipal autonomy means that local interpretations and organisational routines vary considerably. Some municipalities employ dedicated scheduling officers, while others expect preschool heads to create timetables manually using their pedagogical expertise. As SKR's national digitalisation review notes, such decentralisation contributes to "significant administrative variation" across Sweden's approximately 9,000 preschools [2].

2.1 The Administrative Reality

In practice, staff scheduling is a recurring and labour-intensive administrative activity. Preschool heads often rebuild rolling schedules every four to six weeks to accommodate changes in staff availability, attendance patterns, and legal requirements. Municipal workload studies and digitalisation reports consistently describe scheduling as one of the most time-consuming administrative tasks in preschool leadership. In several municipalities, scheduling accounts for an estimated 20–30% of a preschool leader's administrative workload, particularly during periods of staff turnover or fluctuating attendance [7, 2]. Similarly, national inspections highlight that administrators frequently

experience high cognitive load related to staffing coordination, documentation of decisions, and ensuring compliance with work environment regulations [8, 7].

Schedulers must continuously balance competing demands: aligning staff coverage with variable child attendance, distributing planning time, respecting contractual limits, and accommodating individual preferences or fairness expectations. Many educators have recurring requests for example, alternating early and late duties or securing one afternoon per cycle for study or family commitments. Each small adjustment can ripple through dozens of interlinked shifts, creating new conflicts or compliance issues. As a result, scheduling becomes a continuous act of negotiation rather than a purely technical puzzle.

Despite the introduction of digital administration systems, most scheduling still occurs in spreadsheets or locally developed templates. Research on workforce rostering shows that many public-sector organisations remain dependent on manual coordination and tacit expertise rather than integrated optimisation tools [3, 4]. In Swedish preschools, human schedulers rely heavily on experiential knowledge who works well together, which educators require mentoring, or which children need consistent support. While this situated knowledge is invaluable, it limits scalability and traceability. When external auditors or unions request documentation, many preschools struggle to reconstruct the rationale behind specific scheduling decisions [2]. Scheduling thus becomes not only administratively costly but also a potential compliance risk.

Empirical studies further show that preschool operations are characterised by high variability and uncertainty. Child attendance patterns fluctuate with illness, seasonal effects, and flexible parental leave. Substitute staff availability is often constrained, particularly in rural municipalities. Pedagogical guidelines emphasise maintaining stable groups and relationships, which can conflict with the flexibility needed to absorb short-notice changes [9, 10, 11]. Interviews conducted during this study revealed that many preschools construct schedules a full term in advance, requiring administrators to predict attendance patterns and staffing needs long before actual child presence is known. Variations in parental work hours, new enrolments, and absences frequently disrupt carefully balanced rosters. These factors collectively make preschool scheduling a prime example of bounded rationality, where decision makers must navigate complex interdependencies under limited time, incomplete information, and

changing conditions.

2.2 Digital Platforms and Their Gaps

Over the last decade, many Swedish municipalities have adopted cloud-based administration systems such as *Tempus*, *Tieto Education*, *Vklass*, and *IST Administration*. These platforms handle time reporting, attendance registration, and communication with guardians. Among them, *Tempus* is one of the most widely used solutions, offering modules for attendance tracking, staff presence, and resource management [5, 12, 13]. However, while *Tempus Admin* provides structured data on child attendance and staff assignments, it still relies on manual configuration to produce coherent weekly plans.

As shown in Figure 1, the existing *Tempus Admin* platform provides a digital overview of staff and child attendance but offers limited support for automated rostering. This environment served as the technical and organisational foundation for the smart scheduling function developed in this project.

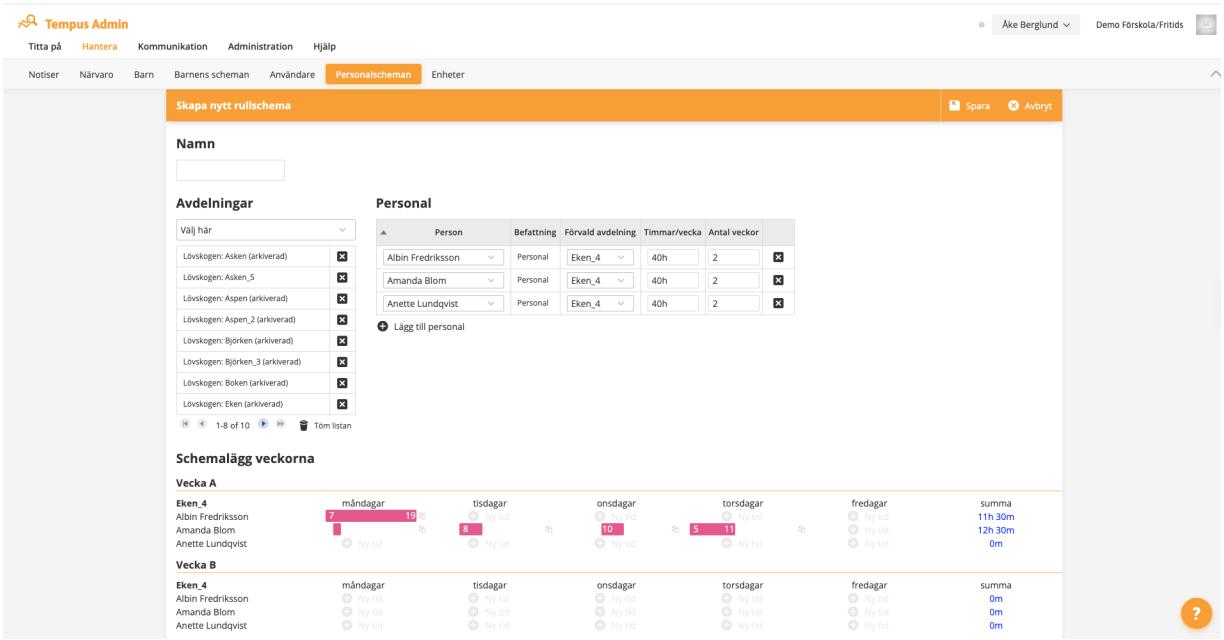


Figure 1 – Current interface of the *Tempus Admin* system, which served as the integration environment for the smart scheduling prototype.

This project addresses that gap by developing a smart scheduling function embedded within *Tempus Admin*. The goal is to combine existing operational data with configurable rules to generate draft schedules that are legally compliant, pedagogically balanced, and easily adjustable. By surfacing explanations

and trade-offs, the function aims to transform scheduling from a repetitive manual task into an interactive decision-support process. Similar approaches in other sectors (e.g., healthcare, retail) have demonstrated improvements in both schedule quality and user satisfaction when algorithmic tools are used transparently [4, 14, 15, 16, 17].

2.3 Algorithmic Reasoning for Scheduling

Algorithmic reasoning for staff scheduling can be broadly understood as the use of explicit rules, constraints, and optimisation principles to generate or evaluate possible timetables. In contrast to machine-learning-based prediction, rule-based scheduling represents domain knowledge in a transparent and inspectable form. This tradition has long been central to knowledge-based systems and algorithmic decision-support tools, where the goal is not autonomous decision-making but structured reasoning that aligns with formal and contextual requirements [18].

In public-sector domains such as healthcare, transportation, and workforce management, constraint-based scheduling has proven effective for balancing legal conditions, operational goals, and fairness-oriented preferences [3, 4]. These systems typically distinguish between *hard constraints*, which capture non-negotiable requirements such as working-time regulations or safety standards, and *soft constraints*, which encode desirable properties such as fairness, stability, or individual preferences.

For preschool scheduling, this approach aligns well with the normative and compliance-oriented nature of the domain. Legal requirements, pedagogical principles, and collective agreements can be formalised as explicit rules, while local practices and fairness norms can be modelled as adjustable preferences. Because transparency and accountability are central in public administration, rule-based methods offer significant advantages over black-box predictive systems: stakeholders can inspect the logic, understand why particular assignments are made, and contest or override decisions when necessary.

Algorithmic reasoning in this context therefore serves as a form of decision support rather than automation. Schedulers remain responsible for interpreting recommendations, resolving conflicts, and applying professional judgement. This emphasis on human oversight reflects broader frameworks for responsible

and value-aligned automation in the public sector [19, 20, 21, 22].

2.4 From Scheduling Theory to Practice

From a computational standpoint, preschool scheduling can be approached as a constrained optimisation problem balancing strict rules with human preferences. Hard constraints capture non-negotiable legal and operational requirements such as daily and weekly rest periods, staff–child ratios, and maximum working hours, while soft objectives represent desirable conditions, including fairness in shift rotation, pedagogical continuity, and workload balance. These objectives often pull in different directions improving one aspect, such as fairness, may worsen another, such as stability or coverage. Rather than producing a single "best" roster, the algorithm aims to generate balanced solutions that make such trade-offs visible and adjustable to human schedulers.

However, transferring these techniques to the preschool domain introduces distinct challenges. Unlike hospital or airline rostering, where quantitative metrics dominate, preschool scheduling involves normative and relational dimensions: educators' wellbeing, fairness across weeks, and children's security. These are difficult to capture in numerical objectives yet central to perceived quality. Consequently, the project adopts a hybrid approach, combining rule-based reasoning with lightweight optimisation, ensuring that outputs remain interpretable and adjustable by users.

2.5 Human–Algorithm Collaboration and Cognitive Load

Designing such systems requires careful attention to human cognitive limits and interaction patterns. Classic usability research shows that overly complex or opaque automation can overwhelm users and erode trust [23, 24]. To address this, the project follows the Double Diamond design framework [25], progressing through discovery, definition, development, and delivery in close collaboration with real schedulers. Participatory workshops help uncover how users articulate constraints, resolve conflicts, and perceive fairness in practice.

Research on explainability and human–computer interaction emphasises the importance of transparency and user control. Users are more likely to trust algorithmic recommendations when they can understand the reasoning behind them, see which constraints influenced each decision, and easily override re-

sults if necessary [26, 27, 20]. In this project, explanations are communicated through multiple channels: visual indicators highlighting conflicts or rule compliance, concise textual rationales (e.g., "Assigned late shift to maintain rotation fairness"), and interactive panels where users can review and validate system suggestions. Usability feedback from iterative testing sessions informed the ongoing refinement of both interface and logic.

By combining explainability with user control, the system aims to augment rather than replace human judgement. This aligns with participatory design traditions and emerging literature on human oversight in algorithmic decision making [19]. The result is an interactive environment where users can iteratively refine generated schedules while maintaining a clear mental model of how assignments were produced.

2.6 Responsible and Compliant Automation

Because preschool scheduling directly affects working conditions, the system qualifies as a high-risk automated decision-support tool under the EU Artificial Intelligence Act (2024/1689) [28]. Compliance requires documentation, risk management, human oversight, and transparency. To address these requirements, the project includes mechanisms for model cards, traceable decision logs, and review-and-approve workflows. In accordance with the General Data Protection Regulation (GDPR), the project applies data minimisation, access control, and anonymisation principles [29]. During prototyping, synthetic datasets are used to avoid exposure of personal data.

Accessibility is treated as a first-class requirement, guided by WCAG 2.2 standards [30]. This ensures that the interface remains usable for administrators with varying digital literacy and abilities. Furthermore, auditability is embedded so users can export explanations and constraint reports to demonstrate compliance to internal auditors or external authorities such as the Work Environment Agency.

2.7 Cultural and Organisational Context of Preschool Scheduling in Sweden

While legal and pedagogical regulations provide a formal structure for preschool operations, scheduling practices in Sweden are also deeply shaped by cultural,

organisational, and practical factors [8, 2]. Interviews conducted for this study confirmed that many decisions are guided by local routines, tacit knowledge, and professional norms rather than written rules. Understanding these everyday realities is essential for designing algorithmic tools that reflect not only compliance but also lived professional practices.

Preschools in Sweden typically operate from early morning (around 06:30) until late afternoon (17:30–18:00), providing full-day care and education for children aged one to six. Within these long opening hours, activities follow a pedagogical rhythm that includes play, learning sessions, outdoor time, and rest. Staff schedules must align with these temporal anchors, ensuring that qualified preschool teachers lead pedagogical activities during designated "teaching hours", while childcarers support during broader care periods [1]. This structure introduces temporal constraints distinct from those in most industrial scheduling domains: the day cannot be partitioned arbitrarily but must respect pedagogical sequences and team constellations.

Operational dependencies across municipal services add further complexity. In many municipalities, preschool meals are not prepared on-site but supplied through central kitchens or outsourced catering contracts [2, 31]. Mealtimes are therefore externally determined typically around 11:00 for lunch and 14:00 for afternoon snacks and cannot be easily shifted without breaching supplier schedules. Interview participants noted that these fixed service windows often constrain pedagogical flexibility and staff planning as supervision must be ensured during deliveries and meals. Similarly, cleaning or maintenance services may operate on separate timetables, requiring staff to vacate certain spaces or complete activities before end-of-day routines. These dependencies illustrate how scheduling extends beyond the preschool itself, forming part of a broader municipal logistics system.

Cultural norms also play a crucial role in shaping scheduling preferences. Swedish preschools operate with a strong ethos of collegiality and equality, reflecting broader societal values of fairness and work–life balance. Since public-sector preschools rarely offer financial bonuses or flexible benefit systems, shift allocation often functions as a form of compensatory fairness. Rotating early and late shifts allows staff to share less desirable working times equitably. In some units, it is an established local norm that every fifth Friday or similar in-

terval grants staff an earlier departure, a small but meaningful gesture that reinforces perceived justice and morale. These informal arrangements are rarely codified in regulations but are crucial for maintaining trust within teams [8, 2].

At the same time, these cultural practices create implicit constraints that algorithmic tools must handle sensitively. A mathematically "fair" rotation may still feel unjust if it disrupts long-standing routines or informal agreements. Conversely, respecting tacit practices may improve acceptance even if strict formal fairness is not achieved. Thus, culturally informed scheduling requires not only constraint satisfaction but also awareness of the social fabric in which decisions are embedded. Capturing such soft factors poses a methodological challenge for smart scheduling systems, underscoring the need for participatory design and explainability in implementation.

3 Design Frameworks Guiding the Study

The methodological foundation of this study combines three established frameworks from design research and human–computer interaction: Design Science Research, the Double Diamond model, and Participatory Design. Each framework contributed a complementary perspective on how to structure, conduct, and evaluate the design of the algorithmic scheduling function for Swedish preschools.

Figure 2 provides an overview of how these frameworks were integrated in practice. The Double Diamond structured the overall research flow from discovery to delivery, while Design Science Research guided the iterative *build–evaluate–reflect* cycles applied in each design iteration. Participatory Design principles underpinned all stages of the process, ensuring user engagement and contextual relevance.

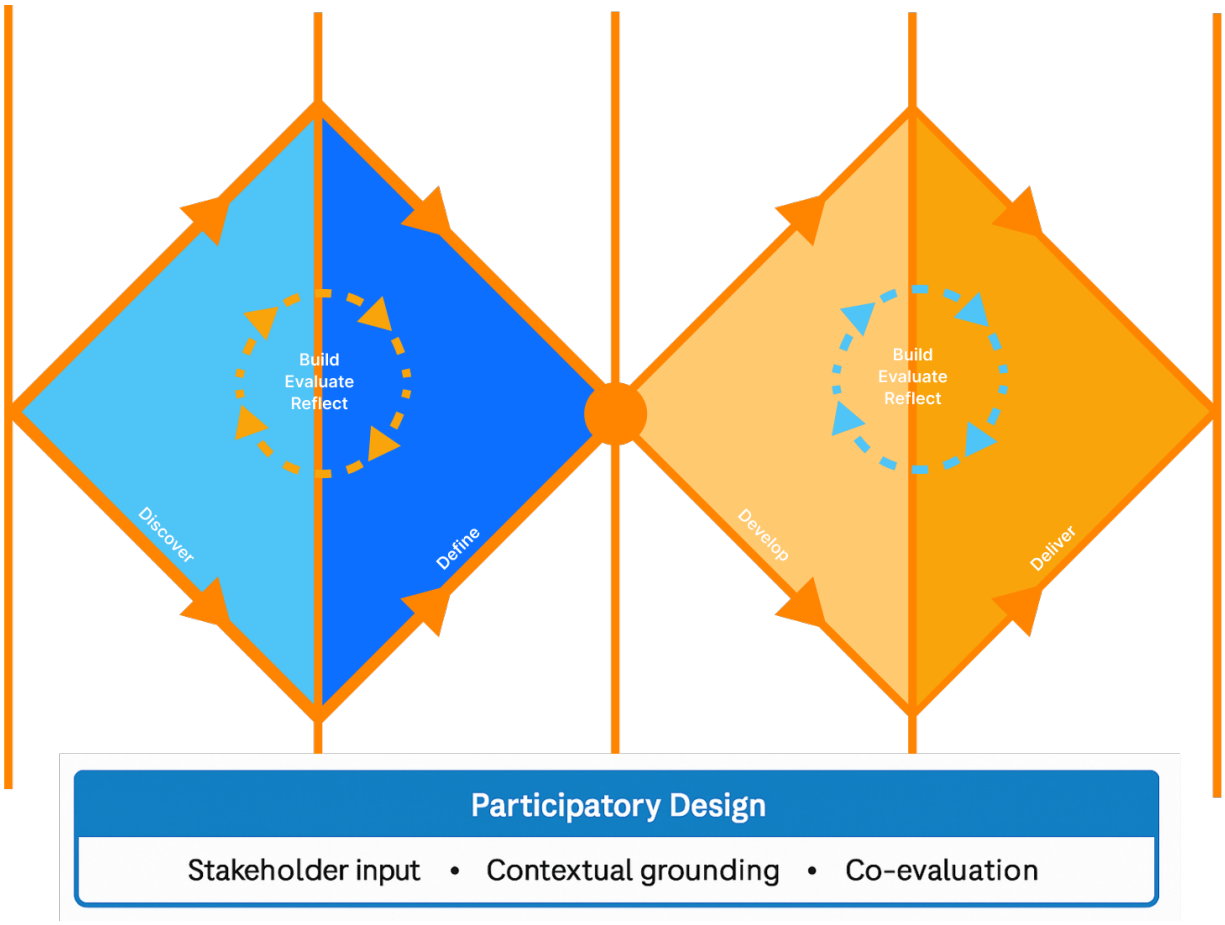


Figure 2 – Integration of design frameworks guiding the study. The Double Diamond model structured the overall process, Design Science Research defined the iterative build–evaluate–reflect cycles within each phase, and Participatory Design principles ensured continuous stakeholder involvement. Figure illustrated by the author using Figma.

3.1 Design Science Research (DSR)

Design Science Research (DSR) provides a systematic approach for creating and evaluating artefacts intended to solve real-world problems while generating transferable knowledge [32, 33]. In DSR, knowledge is produced through the design and evaluation of an artefact that addresses an identified problem within a specific context. The framework typically follows iterative cycles of *build*, *evaluate*, and *reflect*, aligning well with the technological and applied nature of this project.

In this study, the artefact was a functional prototype of an algorithmic scheduling system designed for integration within *Tempus Admin*. Each iteration of the research cycle contributed both practical improvements to the system

and conceptual understanding about human–algorithm collaboration in public-sector decision support. Following DSR ensured that technical development and theoretical reflection advanced together, connecting the artefact's performance and user experience to broader questions about explainability, trust, and fairness in algorithmic systems.

In line with DSR principles, both qualitative and limited quantitative feedback were collected during evaluation phases, including short Likert-scale items [34] to complement interview data with structured user ratings of fairness, usability, and explainability.

3.2 Double Diamond Model

The Double Diamond model, developed by the UK Design Council, conceptualises innovation as a sequence of divergent and convergent phases: *Discover*, *Define*, *Develop*, and *Deliver* [25]. It emphasises exploration before solution, encouraging designers to first broaden their understanding of the problem space before narrowing down toward implementation.

This framework structured the progression of the study into distinct yet connected stages. The *Discover* and *Define* phases corresponded to Iteration 1, where the problem domain preschool scheduling practices was explored through contextual interviews and document analysis. The *Develop* and *Deliver* phases informed Iterations 2 and 3, which focused on prototyping, user feedback, and evaluation. Visualising the process through the Double Diamond helped communicate the iterative, reflective nature of the design process to participating stakeholders and ensured a balanced focus on both problem understanding and solution testing.

3.3 Participatory Design (PD)

Participatory Design (PD) is a Scandinavian tradition in design research that emphasises the active involvement of users in shaping the systems they will ultimately use [35, 36]. It recognises users as experts in their own work practices and views design as a democratic and collaborative process.

This study adopted PD principles to engage preschool administrators, municipal representatives, and developers from *Tempus Information Systems AB* as

co-creators throughout the project. Participants contributed not only feedback on interface prototypes but also helped define design requirements, fairness principles, and explanatory features. Through iterative workshops and prototype sessions, users' insights directly influenced how the algorithmic scheduling function handled rotation rules, coverage balance, and pedagogical constraints. This participatory approach ensured that the resulting design was grounded in real-world needs and that the emerging algorithmic functionality supported rather than replaced professional judgement.

3.4 Qualitative and Semi-Structured Methods

In addition to the design-oriented frameworks, the study employed qualitative research methods to capture the situated experiences and reasoning of preschool administrators. Semi-structured interviews were chosen as the primary qualitative data collection technique because they combine the comparability of a structured format with the openness needed to explore participants' lived practices and contextual nuances [37]. This approach aligns with participatory and user-centred traditions, which emphasise understanding users' perspectives and integrating their experiential knowledge into the design process.

4 Methodology

This chapter describes the research methodology used to design, develop, and evaluate the smart scheduling function within *Tempus Admin*. The project follows a mixed-method, design-oriented approach grounded in human–computer interaction (HCI), design science research, and participatory design principles. The methodology combines iterative prototyping with empirical user feedback to ensure both technical validity and contextual fit within Swedish preschool administration.

4.1 Research Design

The overall research strategy followed a *design–build–evaluate* cycle consistent with design science research [32]. The goal was not only to construct a functional prototype but also to generate transferable knowledge about how smart can support scheduling in public-sector contexts. Each iteration involved problem analysis, prototype design, implementation, and evaluation, guided by the Double Diamond framework for innovation [25].

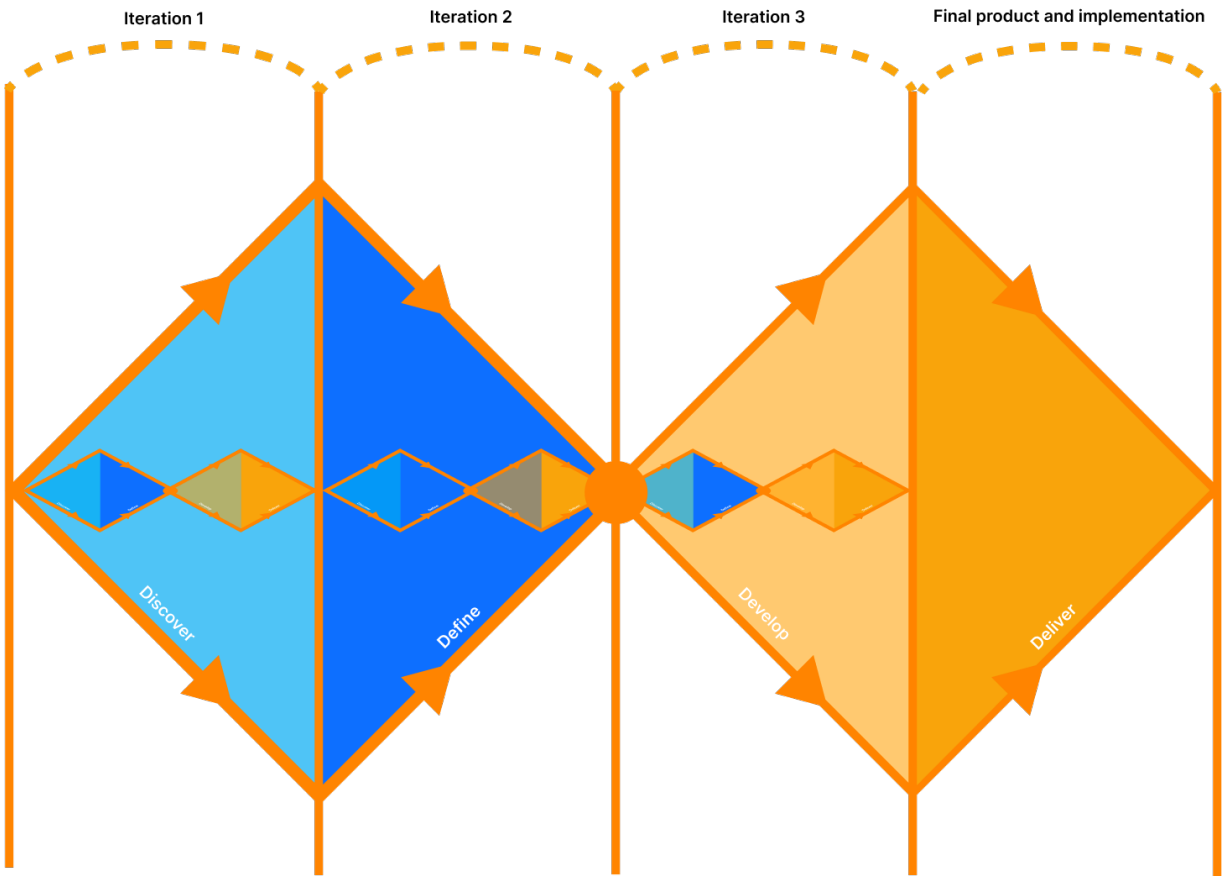


Figure 3 – Iterative application of the Double Diamond model across three design cycles. Each iteration included discovery, definition, development, and delivery activities. Figure illustrated by the author using Figma.

During the *Discover* phase, existing scheduling practices were mapped through document review and exploratory interviews with preschool leaders and municipal administrators. The *Define* phase synthesised these insights into design requirements and constraints, such as coverage rules, pedagogical structure, and fairness considerations. In the *Develop* phase, a series of prototypes were implemented and tested. Finally, in the *Deliver* phase, a validated prototype was integrated into a sandbox version of *Tempus Admin* for empirical user evaluation.

This approach allowed the study to combine technical research (optimisation modelling, constraint encoding, system integration) with applied HCI research (interaction design, explainability, user trust). The method also acknowledged the sociotechnical nature of the problem where technology and practice co-evolve through iterative engagement [38, 39]. Figure 3 shows how the Double Diamond model was applied iteratively across the three design cycles.

4.2 Iteration Cycles

The development of the scheduling prototype followed three iterative cycles, each building upon insights from the previous phase. The process was guided by the Double Diamond model of *Discover–Define–Develop–Deliver*, ensuring that user needs, design activities, and technical refinement evolved in a structured and traceable way. Each iteration served a specific purpose within the broader research design and was informed by methodological principles of participatory and user-centred development. Table 1 outlines the purpose, activities, and outcomes of each design iteration.

Table 1 – Overview of the three design iterations conducted in the study. Each iteration built upon the previous one following the Double Diamond process.

Iteration	**Purpose**	**Main Activities**	**Outcome / Focus for Next Step**
1: Understanding Context	Explore current scheduling practices, challenges, and constraints in Swedish preschools.	Semi-structured interviews with preschool administrators and analysis of existing documentation.	Identification of key challenges and initial design requirements.
2: Prototype Development and Feedback	Translate insights from Iteration 1 into a functional prototype and collect user feedback on logic, usability, and fairness mechanisms.	Participatory prototype workshops in four municipalities (*Örebro, Stockholm, Helsingborg, Umeå*).	Refinement of the prototype and definition of focus areas for Iteration 3.
3: Evaluation and Refinement	Assess refined prototype variants and investigate user perceptions of realism, fairness, and explainability.	Comparative user testing with different constraint configurations and follow-up discussions.	Formulation of final design principles for explainable, configurable scheduling support.

Table 2 summarises how different municipalities contributed data across the three design iterations.

Table 2 – Overview of study iterations, data collection methods, number of workshops or interviews, participant involvement, and represented municipalitiesß.

Iteration	**Data sources**	**Municipalities involved**
Iteration 1	Contextual interviews and document analysis	3 semi-structured contextual interviews with 7 participants (Helsingborg), supplemented by document analysis of procurement and policy documents from Stockholm, Haninge, Varberg, Örebro, and Jönköping
Iteration 2	Participatory prototype workshops	5 prototype evaluation workshops with 7 returning participants from Iteration 1, together with 5 additional participants (1 from Stockholm, 1 from Örebro, and 3 from Umeå)
Iteration 3	Evaluation workshops	4 follow-up evaluation workshops with all participants from Iteration 2 except the participant from Örebro, who was unable to attend due to scheduling conflicts

The first iteration focused on understanding the context in which preschool scheduling occurs and identifying requirements for digital support. To achieve this, semi-structured interviews were conducted in Helsingborg, complemented by document analysis from five additional Swedish municipalities. Participants included preschool heads, administrators, and digitalisation coordinators, representing different organisational sizes and governance models. Supporting materials such as procurement documents, requirement lists, and internal policy guidelines were collected to capture both practical and institutional perspectives. Each interview lasted approximately one hour and followed a thematic protocol covering current scheduling routines, challenges, and expectations for automation. All interviews were recorded and transcribed for subsequent thematic analysis. Sessions were primarily conducted online, with the exception of those held in Umeå, which took place in person. The data gathered during this phase informed the definition of user requirements and constraint parameters to guide the first prototype.

Building on these contextual insights, the second iteration involved the cre-

ation and evaluation of a proof-of-concept prototype integrated into a sandbox version of *Tempus Admin*. This prototype allowed participants to explore features such as constraint configuration, roster generation, and visual feedback on compliance. Participatory workshops were conducted in four municipalities *Örebro, Stockholm, Helsingborg, and Umeå* to capture diverse operational contexts. Each session included between one and three participants and lasted around one hour. Participants were asked to interact with the prototype while thinking aloud, focusing on aspects such as clarity of logic, usability, and alignment with everyday scheduling practices. Observations, screen recordings, and post-session reflections were used to document user interaction and identify potential improvements. This iteration primarily aimed to assess procedural usability and gather feedback on design features rather than to evaluate effectiveness or outcomes.

The third iteration introduced refined prototype versions incorporating new rule configurations and explainability mechanisms derived from earlier workshops. The evaluation adopted a comparative format, presenting participants with several automatically generated schedules based on different constraint strategies: coverage-oriented, rotation-based, and hybrid adaptive. Participants were invited to compare and discuss these schedules in terms of interpretability, plausibility, and perceived fairness. Each session followed a semi-structured discussion format, enabling participants to articulate their reasoning processes and expectations for transparency. These activities provided structured input for assessing how different algorithmic strategies aligned with professional judgement and organisational values in preschool scheduling practice.

To complement these qualitative discussions, short post-session questionnaires using five-point Likert scales were introduced. These items captured participants' quantitative ratings of perceived fairness, usability, and transparency in the generated schedules. This mixed-method approach allowed the study to triangulate subjective impressions with structured comparisons across participants.

Overall, the three iteration cycles operationalised a participatory design approach where stakeholder engagement, prototype development, and empirical evaluation were tightly integrated. This iterative structure ensured that

methodological rigour was maintained while allowing the artefact to evolve responsively to user feedback and contextual constraints. Findings and implications from each cycle are presented in detail in Chapter 5.

4.3 User Involvement and Data Collection

Participants were recruited in collaboration with municipal partners and *Tempus Information Systems AB*. All participants held managerial or administrative responsibility for preschool operations primarily preschool heads and administrative staff responsible for staff rosters, attendance planning, or system configuration. Their experience provided insight into the intersection of pedagogical goals, legal rules, and local working culture.

In total, fifteen individuals participated across all iterations. Seven participants contributed during the first iteration through contextual interviews, while eight participated in prototype evaluations during the second and third iterations. The municipalities represented included Stockholm, Helsingborg, Örebro, Umeå, Haninge, Varberg, and Jönköping. Participation varied by iteration and data type, as detailed in Table 2. This ensured diversity in operational scale, organisational structure, and digital maturity.

In later evaluation sessions, semi-structured interviews were complemented with brief Likert-scale questionnaires [34]. These scales provided a simple quantitative complement to the qualitative data, helping identify patterns in user perceptions of fairness, usability, and system transparency across different prototype versions [40].

Interview Structure and Procedure. All interviews and prototype sessions followed a semi-structured format designed to balance comparability across participants with flexibility for in-depth exploration. Each interview began with questions about the participant's professional role, scheduling responsibilities, and years of experience. This was followed by discussions of daily workflows, decision-making routines, and practical challenges. Sessions concluded with reflections on the idea of algorithmic decision support, expectations regarding fairness and transparency, and thoughts on usability and control in digital scheduling systems.

All interviews were conducted in Swedish the participants' first language and

took place either online via Microsoft Teams or in municipal offices, depending on participant preference. Each session lasted approximately 45–60 minutes. With participants' informed consent, sessions were audio recorded and later transcribed verbatim. Transcripts were translated into English for analysis while retaining key Swedish terms where direct equivalents were lacking. Summarised transcripts were returned to participants for member checking to verify accuracy and interpretation.

During the prototype sessions, participants were encouraged to demonstrate their existing scheduling workflow by sharing their screens. This enabled direct observation of how schedules were constructed, adjusted, and evaluated in real time, revealing tacit reasoning and recurring obstacles. Field notes and annotated screenshots complemented the recordings, providing a detailed picture of user behaviour and contextual factors.

Rationale for Using Semi-Structured Interviews. Semi-structured interviews were selected because they allow participants to articulate complex, context dependent experiences while maintaining enough structure to enable thematic comparison across cases [37, 41]. Given the exploratory and participatory character of this study, this format supported open-ended discussion of scheduling practices, fairness norms, and expectations for algorithmic support while ensuring alignment with the study's guiding research questions.

Data collection combined several qualitative techniques:

- **Contextual interviews** explored existing scheduling workflows, recurrent challenges, and fairness norms such as rotation practices or informal compensations.
- **Participatory prototype sessions** enabled participants to interact with evolving design concepts, evaluate usability, and discuss explainability and fairness in generated schedules.
- **Reflective discussions** followed each prototype session to capture participants' interpretations of automation and their perceived trust or concerns.

See the full interview guides in Appendices A and B. All data were analysed thematically, focusing on patterns related to usability, fairness, explainability,

and organisational fit. Coding was conducted iteratively between design cycles to inform subsequent prototype refinements.

4.4 Prototype Development

The scheduling prototype was developed in close collaboration with *Tempus Information Systems AB*, which provided access to an anonymised test environment and representative datasets. Before implementation, early wireframes were created in Figma [42] to explore layout, terminology, and user interactions for defining rules and constraints. These low-fidelity mock-ups supported participatory feedback during the *design–build–evaluate cycle* (Figure 4).

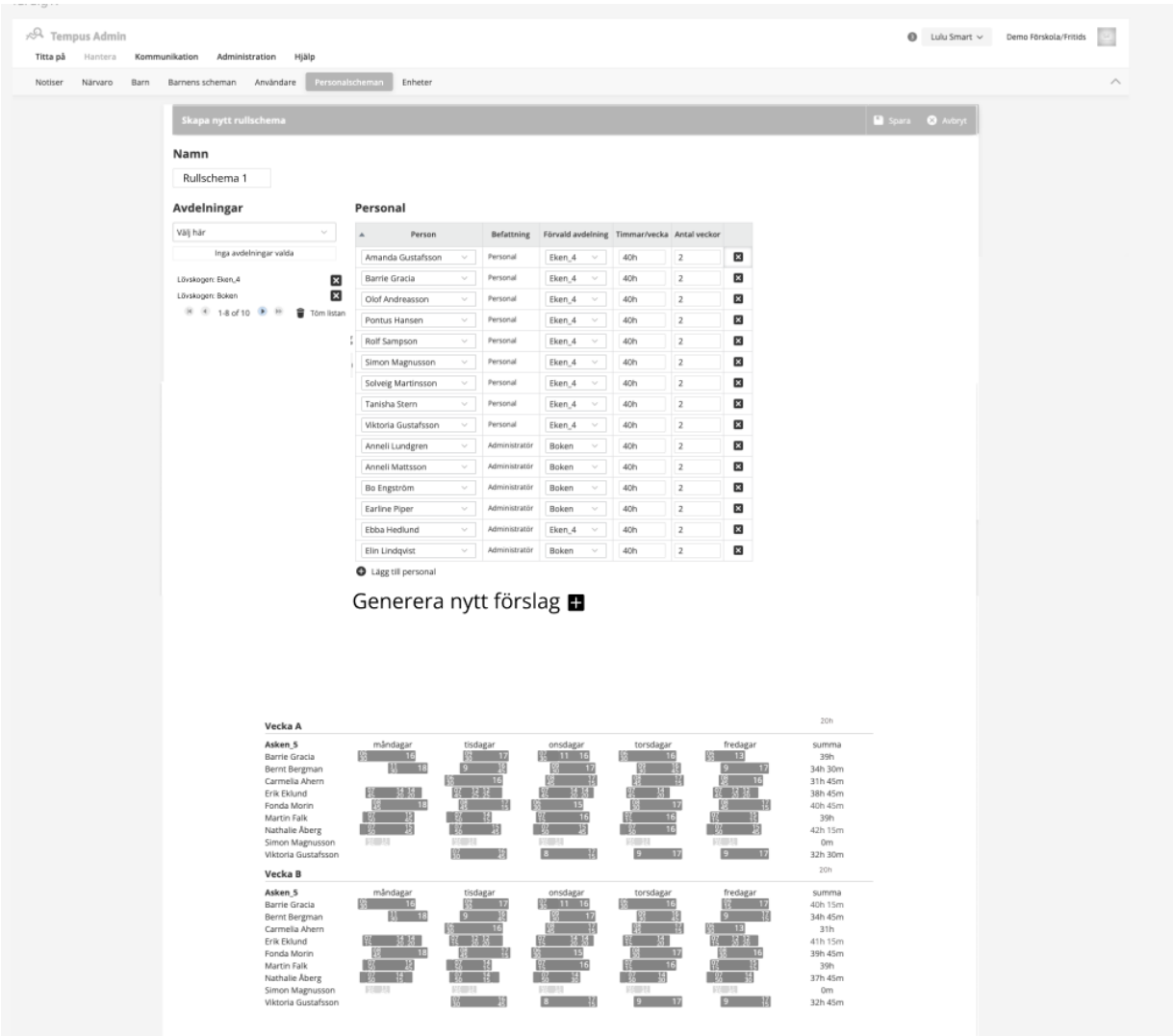


Figure 4 – Early design proposal created in Figma to explore layout, user flow, and rule-configuration concepts prior to implementation.

The technical implementation followed a three-layer architecture:

1. **Constraint modelling.** Legal, organisational, and pedagogical rules were encoded as hard and soft constraints using a constraint programming (CP) framework, including rest rules, staffing ratios, role balance, and fairness rotations.

2. **Optimisation and reasoning.** A hybrid CP heuristic engine generated feasible rosters balancing objectives for coverage, fairness, and stability. Adjustable weights allowed experimentation with different value priorities.

3. **Explainable interface.** The user interface displayed generated rosters, constraint violations, and rationales (e.g., "assigned late shift to maintain rotation equity"). A visual timeline view supported interactive edits and counterfactual exploration.

The prototype was implemented in Python with a lightweight web interface built in React, enabling rapid iteration. Integration with the *Tempus Admin* system allowed realistic data import and export. Weekly feedback sessions with *Tempus* designers and municipal stakeholders ensured alignment with operational requirements.

Rationale for Choosing Python and OR-Tools. Python was chosen for its rapid prototyping capabilities and its mature ecosystem for optimisation and constraint modelling. The project relied on Google's OR-Tools [43], recommended by engineers at *Tempus Information Systems AB* as an effective framework for exploring scheduling logic and one they might be interesting using for later development. OR-Tools provides efficient solvers for constraint programming, mixed-integer optimisation, and routing problems, enabling a transparent and modular representation of pedagogical, organisational, and legal rules.

The combination of Python and OR-Tools supported fast iteration throughout the *design–build–evaluate* cycle: constraints could be adjusted between workshops, and different scheduling strategies generated without major code restructuring. This offered a balanced mix of development speed, solver performance, and methodological fit for exploratory design science research.

4.5 Evaluation Procedure for Iteration 3

The third design iteration required a structured evaluation procedure to compare three different scheduling strategies and to assess perceptions of fairness, predictability, and applicability in real preschool settings. This subsection describes the evaluation setup, presentation method, and data collection process used during Iteration 3.

Parameter Review and Configuration. Each evaluation session began with a walkthrough of the prototype's configuration interface, where participants reviewed the parameters used to generate multi-week schedules. These included coverage requirements, opening and closing rules, staff availability, merged operations, and additional constraint settings. This step ensured that participants understood the underlying assumptions before examining the generated schedules.

Presentation of Schedules in a Familiar Interface. To maintain ecological validity, automatically generated schedules were imported into the existing municipal scheduling interface used by participants in their daily work. Schedules appeared as if generated "at the press of a button," allowing participants to focus on the realism and practical usability of the results rather than prototype imperfections.

Comparison of the Three Scheduling Strategies. Participants examined three alternative generation strategies: a coverage-first repeated-week approach, a rotation-based approach with local repairs, and a hybrid Friday-cycling approach. The schedules were shown side by side for direct comparison of fairness distribution, stability, and pedagogical alignment. Visual comparison facilitated participants' articulation of trade-offs between regularity and fairness.

Fairness Audit Metrics. A simple fairness audit summarised early starts, late ends, early Fridays, and deviations from proportional weekly hours for each staff member. These metrics supported discussion but were chosen for interpretability rather than formal optimisation.

Structured Likert-Scale Ratings. After reviewing and discussing each strategy, participants rated them using three 5-point Likert scales covering

perceived fairness, likelihood of use, and trustworthiness. Numerical ratings were paired with qualitative explanations collected during the discussion, enabling triangulation between subjective perception and structured comparison.

This evaluation procedure combined parameter inspection, ecologically valid presentation, fairness metrics, and structured scoring to elicit fine-grained feedback on how algorithmic schedules align with practical expectations in Swedish preschools.

4.6 Constraint Model and Algorithmic Approach

The smart scheduling prototype developed in this study is based on explicit, rule-driven reasoning rather than predictive modelling or machine learning. The technical foundation combines constraint programming (CP) with lightweight optimisation techniques to generate feasible and explainable draft schedules. This subsection describes the computational model and algorithmic procedures implemented during the prototype development.

Constraint Representation. Legal, organisational, and pedagogical requirements were encoded as *hard constraints* that must be satisfied for a schedule to be considered valid. These included:

- statutory working-time regulations such as daily and weekly rest periods;
- staffing ratios derived from the Education Act and local policies;
- qualification requirements ensuring that preschool teachers lead pedagogical activities during designated hours;
- maximum shift lengths and rules for breaks.

In addition, *soft constraints* captured desirable but negotiable objectives such as fairness in rotation of early and late shifts, workload balance, stability across weeks, and alignment with pedagogical rhythms. Each soft constraint was associated with a configurable weight, enabling experimentation with multiple prioritisation strategies during evaluation.

Optimisation and Search Strategy. The prototype used a hybrid CP heuristic approach. First, the system constructed an initial feasible roster by satisfying all hard constraints through backtracking search. Next, it explored

alternative schedules by adjusting assignments and evaluating their trade-offs according to the weighted soft constraint objectives. This allowed generation of different scheduling profiles coverage-oriented, fairness-oriented, and hybrid adaptive which were later compared in user workshops.

The search procedure aimed for *balanced* rather than globally optimal solutions. This emphasis aligned with pedagogical and organisational requirements, where transparency, interpretability, and local adjustability are valued more highly than mathematical optimality.

Explainability Mechanisms. To support human oversight, the system produced structured explanations for each assignment and schedule-level outcome. Explanations included:

- visual indicators of constraint satisfaction or violation;
- textual rationales (e.g., "assigned late shift to maintain rotation fairness");
- interactive panels where users could inspect, modify, or override system decisions.

These mechanisms were iteratively refined based on feedback from prototype sessions.

System Integration. The algorithmic engine was implemented in Python and integrated into a lightweight React-based interface. The prototype connected to a sandbox environment of *Tempus Admin*, enabling import of representative child-attendance and staff-availability data. This integration allowed participants to evaluate the system using realistic operational conditions while ensuring compliance with ethical and legal guidelines.

Rationale for Approach. A rule-based, constraint-driven method was selected to ensure transparency, accountability, and ease of human oversight qualities essential in public-sector decision-support systems. Unlike black-box prediction models, the implemented approach enabled clear traceability of each scheduling decision, supporting alignment with the EU AI Act's emphasis on explainability and human control.

Overall, the algorithmic framework served as a decision-support tool that complemented rather than replaced professional judgement, enabling schedulers to review, adjust, and justify system-generated rosters in accordance with local practices and organisational values.

4.7 Ethical and Legal Considerations

Given the project's focus on AI in a public-sector context, ethical compliance and data protection were treated as first-order requirements. The study followed the principles of the EU General Data Protection Regulation (GDPR) [29], applying data minimisation, purpose limitation, and pseudonymisation of all participant data. No personal staff or child information was processed during prototype development; synthetic or anonymised datasets were used.

In alignment with the EU AI Act (2024/1689) [28], which classifies AI systems affecting employment as high-risk, the project incorporated mechanisms for *human oversight*, *traceability*, and *risk documentation*. Participants were informed about the purpose of the study, data handling, and their right to withdraw. Ethical approval for the study was obtained in accordance with institutional guidelines.

Accessibility and inclusion were also considered. Interface elements were evaluated against WCAG 2.2 guidelines [30], and terminology was tested to ensure clarity for non-technical users. These measures align with Swedish public-sector commitments to equality and transparency.

5 Results

This chapter presents the empirical findings generated across the three design iterations. The results illustrate how preschool leaders and municipal administrators interpreted, validated, and challenged the evolving smart scheduling prototype. The structure follows the chronological logic of the design process:

1. Mapping the scheduling context.
2. Developing and refining the prototype through participatory workshops.
3. Evaluating realistic multi-week schedules generated by three alternative strategies.

Across all iterations, results highlight a consistent pattern: users' acceptance of algorithmic scheduling depends less on computational performance and more on whether the system respects professional norms, pedagogical rhythms, and culturally grounded ideas of fairness.

5.1 Iteration 1 – Understanding the Scheduling Context

Iteration 1 aimed to understand how preschool scheduling is currently performed and what challenges practitioners face. The existing *Tempus Admin* interface served as the baseline environment against which the new functionality would eventually be integrated.

Interviews across 6 municipalities revealed that most schedules are created using spreadsheets or custom templates. Participants described the work as a mix of:

- operational constraint solving (coverage, roles, legal requirements),
- informal fairness negotiations (who "deserves" an easier Friday),
- pedagogical planning (keeping teaching teams together),
- interpersonal conflict management.

A typical scheduling cycle involved continuous micro-adjustments due to staff absences, training, sick leave, and fluctuating attendance. The absence of decision-support tools often resulted in hours spent each week adjusting rosters manually. Participants articulated a strong desire for tools that:

- visualise the impact of constraints,
- show fairness distribution over time,
- provide defensible reasoning for staffing decisions.

Differences between municipalities also emerged. Larger cities described more formalised processes, with central Human Resources units or digitalisation coordinators supporting preschool heads, while smaller municipalities relied more heavily on individual principals' initiative. Regardless of size, however, participants agreed that current systems mainly function as "digital paper" rather than active decision-support tools.

These insights informed the initial prototype design for Iteration 2, particularly the decision to focus on visual explanations and clear constraint feedback rather than purely on algorithmic performance.

5.2 Iteration 2 – Prototype Development and Feedback

Iteration 2 introduced the first interactive prototype, consisting of three main user-facing components: a configuration interface, a staff input table, and a generated roster with warnings. The goal was to examine how schedulers understood rule configuration, perceived algorithmic output, and interpreted system explanations.

The findings below are organised around the figures and placed between them, as they were shown sequentially during the workshops.

Prototype Configuration Interface

Figure 5 shows the configuration interface used during Iteration 2, where participants explored rule settings related to coverage, fairness, and working-time constraints.

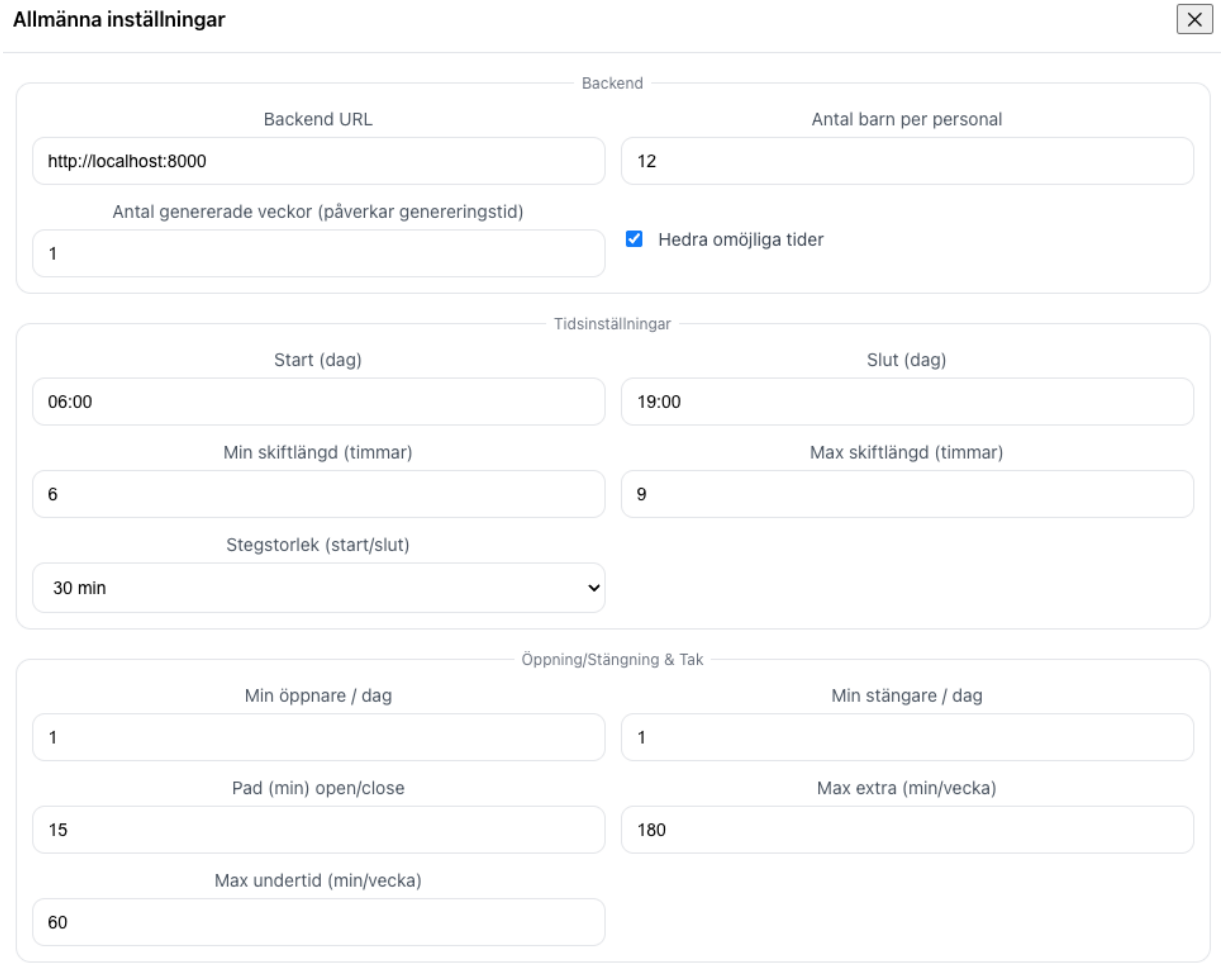


Figure 5 – Prototype configuration interface used in Iteration 2.

Participants explored settings such as:

- shift lengths,
- coverage requirements,
- fairness parameters,
- allowed opening/closing times.

Users appreciated the structured overview but emphasised that real scheduling requires rule exceptions. Several noted that "hard constraints" rarely apply uniformly; for example, planning time for preschool teachers often differs week-to-week. Participants also highlighted the need for defaults that match collective agreements and municipal guidelines, so that new users are not forced to configure everything from scratch.

A recurring comment was that the configuration interface supported reflection on their own practice. By seeing constraints presented explicitly, some participants realised that routines they considered "obvious" were in fact local agreements or habits that might not be shared across units.

Staff Input Table

Figure 6 illustrates the staff input table presented during Iteration 2, which prompted discussions about data preparation, system integration, and error prevention.

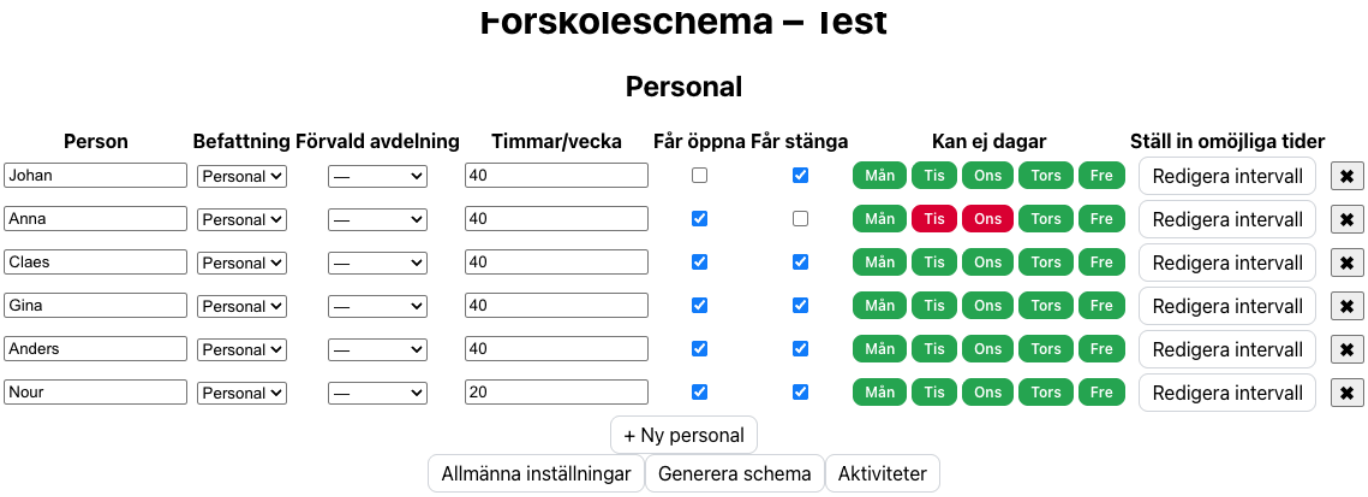


Figure 6 – Staff input interface presented during Iteration 2.

Participants used this interface to reflect on how staff data is currently prepared. Many described spending significant time verifying staff availability and reformatting spreadsheets before they can even begin to think about coverage or fairness. The input table triggered discussions about deeper system integration:

- automatic synchronisation with Human Resources systems,
- pre-population of roles and Full Time Equivalent percentages,
- real-time linking to child attendance patterns.

Several participants pointed out that errors often arise at this early stage: if Full Time Equivalent values or availability windows are entered incorrectly, even a sophisticated scheduling algorithm will produce unsatisfactory results. They therefore saw value in validation mechanisms and clear feedback already at the data-entry level.

Generated Weekly Roster

Figure 7 presents an automatically generated weekly roster, which participants evaluated in terms of feasibility, pedagogical continuity, and workload reduction.

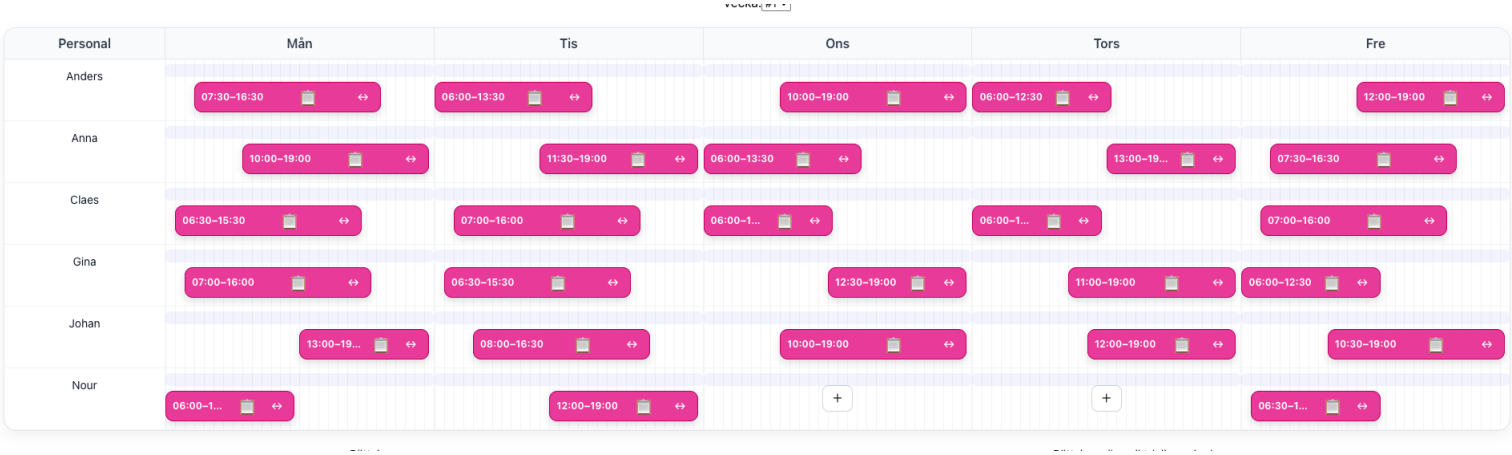


Figure 7 – Generated roster presented during Iteration 2.

Participants examined the automatically generated weekly schedule in detail. They evaluated:

- feasibility of staff allocations,
- respect for planning time,
- distribution of opening/closing duties,
- clarity of the visual layout.

Many appreciated seeing a "complete week appear at once," describing it as a major reduction in workload. However, they also emphasised that the layout must highlight pedagogical anchors clearly, such as:

- who leads each child group,
- continuous assignment of teachers across days,
- identification of unusual or unexpected allocations.

One participant described the generated roster as "a good skeleton, but still in need of flesh," capturing the view that algorithmic output should provide a strong starting point rather than a finished product.

Constraint Violations and Explanations

Figure 8 shows the warning panel highlighting rule violations and explanations, a feature that strongly influenced participants' perceptions of trust and accountability.

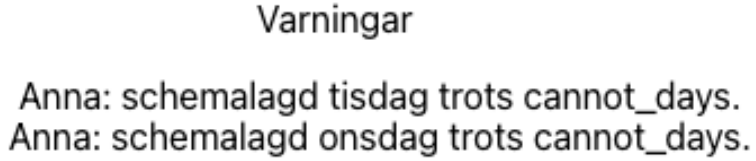


Figure 8 – Warning panel indicating rule violations.

This panel generated the strongest reactions. Participants reported that:

- visible rule violations increased trust,
- explanations reduced the need for manual double-checking,
- warnings felt essential for accountability toward unions and regulators.

One participant said: *"If the system tells me exactly what is wrong and why, then I feel I can stand behind the schedule."*

Others emphasised that warnings also support internal learning: they make it easier to see which local practices systematically collide with legal or organisational constraints. Iteration 2 thus confirmed that transparency and configurability were more valued than optimisation sophistication. This insight shaped the design of the more realistic multi-week evaluation in Iteration 3.

5.3 Iteration 3 – Evaluation and Refinement

In Iteration 3, participants evaluated three alternative scheduling strategies using realistic data and a familiar *Tempus* like interface. To support this evaluation, an updated configuration screen and side-by-side comparison view were introduced. Unlike the previous iteration, the focus was explicitly on the quality of multi-week rolling schedules and how well they aligned with everyday practice, rather than on the novelty of the prototype itself.

Refined Configuration Interface for Iteration 3

Figure 9 shows the refined configuration interface used in Iteration 3 to support multi-week schedule generation and strategic evaluation.

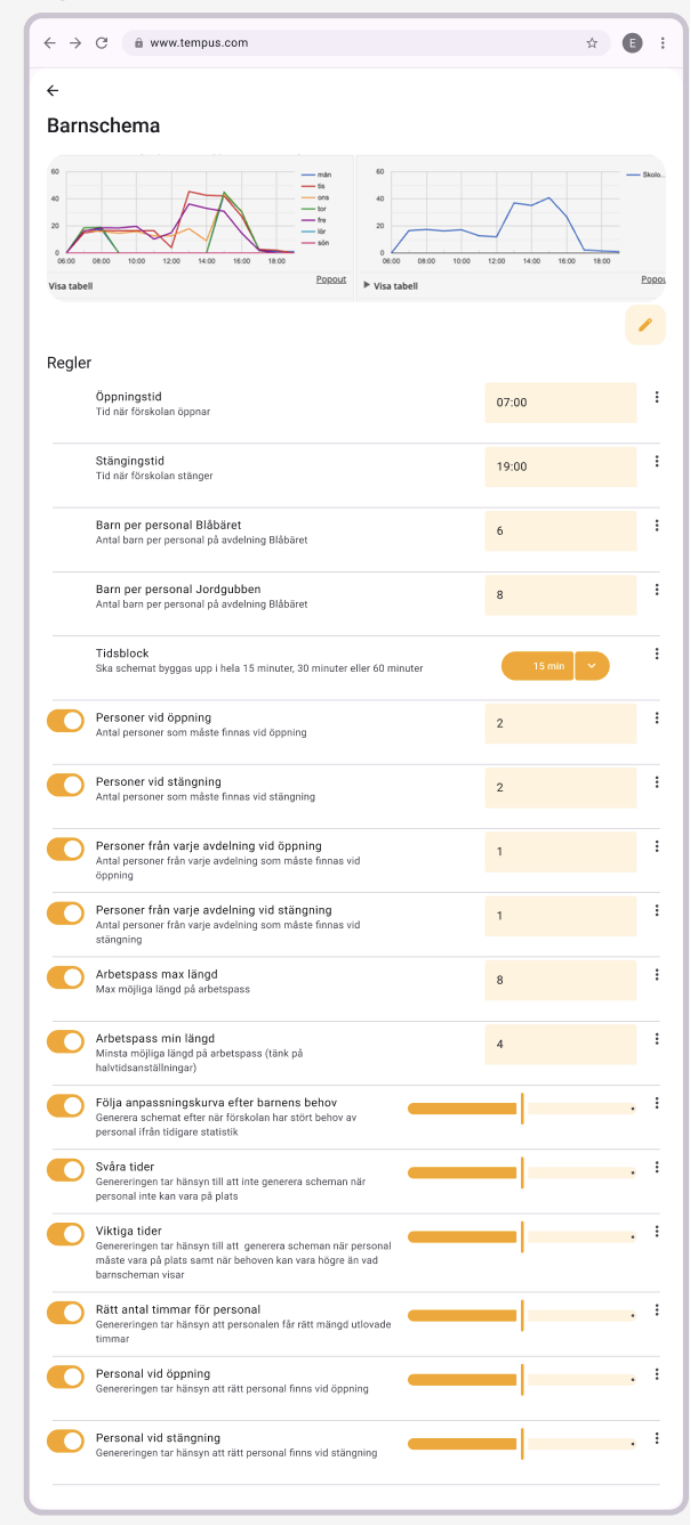


Figure 9 – Refined settings interface used for multi-week schedule generation in Iteration 3.

Participants reviewed attendance curves, opening/closing logic, and role-based constraints. They valued the ability to "see the logic before generating

anything," which increased perceived control and reduced anxiety about pressing the "generate" button. For several participants, this configuration step was also where they negotiated internally which fairness rules and rotation principles should actually be prioritised in their organisation.

Visual Comparison of the Three Strategies

Figure 10 provides a side-by-side comparison of schedules generated by the three alternative strategies, forming the basis for participants' qualitative evaluations.

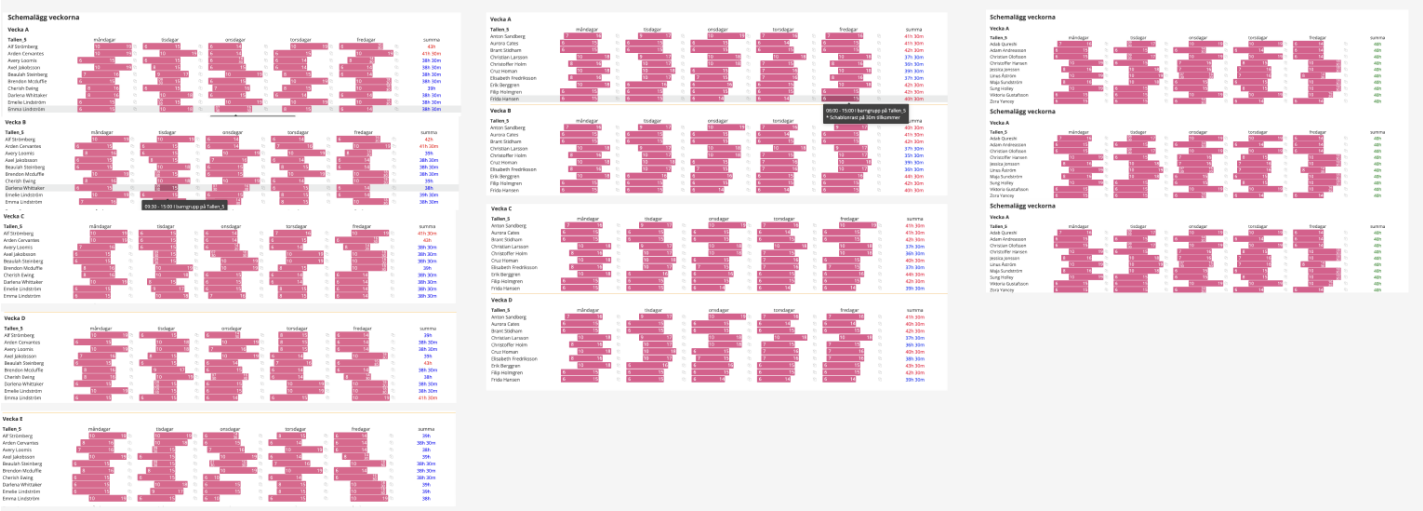


Figure 10 – Side-by-side comparison of the three generation strategies.

This comparison was central to the evaluation. Participants could immediately see differences in:

- fairness distribution,
- week-to-week stability,
- pedagogical continuity.

Rather than discussing algorithms in abstract terms, participants reacted to concrete schedules: who opened, who closed, how often Fridays were "nice", and whether planning blocks appeared in sensible places. The following subsections summarise user reactions to each strategy.

Solution 1 – Perfect Week, Then Repeat

Participants appreciated the predictability of repeating a well-structured week. However, they found this strategy unrealistic:

- It freezes any unfairness present in Week 1.
- Preschool weeks naturally vary; a fixed template does not.
- Staff would quickly perceive patterns as inequitable.

Many described it as useful for inspiration but not adoption. Several suggested that the "perfect week" could be used as a design tool at the beginning of a term, before moving to more nuanced approaches that account for varying attendance and events. Nonetheless, the overall view was that simply repeating a single pattern would generate complaints within a few weeks.

Solution 2 – Rotate and Repair

This strategy was viewed as numerically fair. Across the tested scenarios, early and late duties were spread more evenly across staff, and fairness gaps between individuals were clearly reduced. However, participants raised several concerns:

- Week-to-week variation created stress.
- Children would lose routine anchors.
- Local "repairs" needed clearer explanation.

As one participant said: *"It is fair mathematically, but not humane in practice."* Some municipal coordinators, responsible for ensuring fairness across many preschools, were more positive towards this strategy than individual preschool heads, who focused on day-to-day stability. This tension between central and local perspectives became a recurring theme in the discussions. In terms of predictability, participants explained that the key difference between Solution 1 and Solution 2 lay not in whether patterns existed, but in how visible and stable those patterns were over time. While Solution 1 repeated the same weekly structure, making future assignments easy to anticipate, Solution 2 continuously adjusted individual allocations in response to fairness imbalances. Although these adjustments improved numerical equity, they reduced users' ability to form stable expectations about when people would be working from day to day. Participants therefore described Solution 2 as predictable in aggregate but unpredictable in everyday practice, particularly from the perspective of staff planning their personal routines.

Solution 3 – Hybrid Friday Rotation

This approach stable Monday–Thursday with rotating Fridays was consistently the most accepted. Participants noted:

- It matches informal existing practice.
- Friday is the most flexible day pedagogically.
- Fairness is achieved without destabilising routines.

Many recognised the generated patterns as close to what they were already trying to achieve manually: ensuring that "nice Fridays" are shared fairly without constantly redoing the whole schedule. The strategy was also perceived as easier to explain to staff because the underlying logic (Friday copies a weekday in a cycle) could be communicated succinctly.

5.4 Likert Evaluation of Strategies

To complement the qualitative feedback, participants rated the three strategies on three Likert items (fairness, likelihood of use, and trustworthiness). The aggregated ratings are summarised in Figure 11.

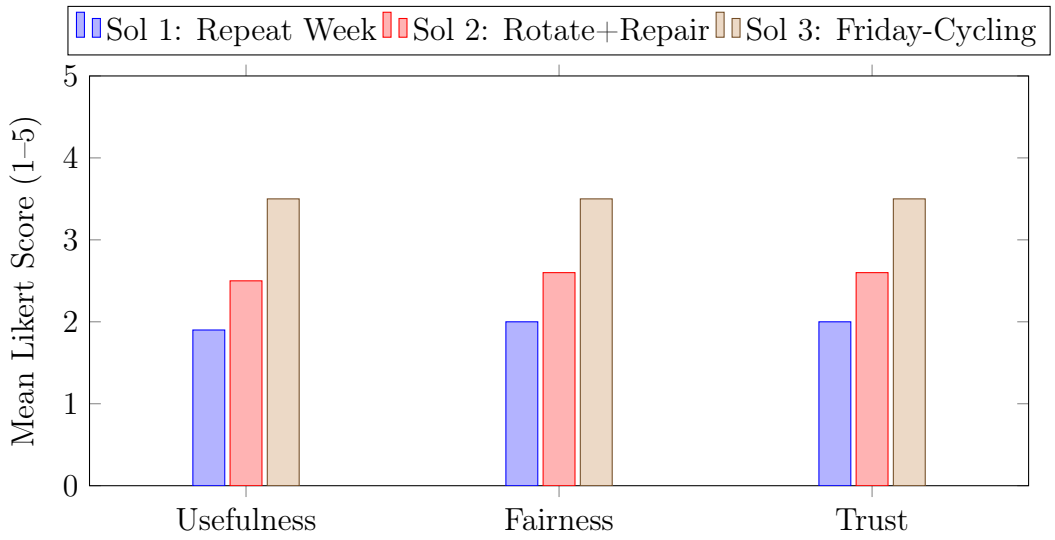


Figure 11 – Mean Likert ratings for the three scheduling strategies. Values represent participant assessments of usefulness, perceived fairness, and trust in real preschool scheduling.

Results indicated:

- Solution 3: highest overall (fairness + usability + trust),

- Solution 2: high fairness, lower predictability,
- Solution 1: high predictability, low fairness.

These numerical patterns reinforced the qualitative impressions: participants were most willing to use the hybrid strategy in real operations, were divided on the rotation-heavy strategy, and saw the repeated-week strategy as mainly a design aid or starting point.

5.5 Summary of results

Thematic patterns emerged early in the study and shaped the direction of subsequent iterations. While the three design cycles had distinct objectives, participants' reflections repeatedly returned to a small set of concerns: administrative workload, fairness and predictability, pedagogical continuity, transparency of system behaviour, and the desire to remain in control of final decisions. The following subthemes were distilled from interviews, workshop discussions, and observations of existing scheduling work.

5.5.1 Administrative Burden and Informal Expertise

Participants emphasised that scheduling is time-consuming and cognitively demanding. Many described the activity as *"a puzzle that only works once you've done it for five years"*. This dependence on tacit expertise often resulted in:

- fragile scheduling practices vulnerable to staff turnover,
- increased emotional labour for schedulers who felt personally responsible for fairness,
- difficulty communicating decisions to staff in a transparent, defensible way.

Several preschool heads described spending entire afternoons or evenings "just moving boxes around" in spreadsheets, often outside regular working hours. When asked what would happen if they changed workplace or went on long-term leave, many admitted that no-one else fully understood their personal shorthand, colour codes, or layout conventions. This sense of being individually responsible, both for the practical work and for keeping peace within the staff group, made the idea of supportive automation attractive but also emotionally charged.

5.5.2 Fairness and Predictability as Cultural Values

Fairness was described not simply as balancing early and late shifts, but as maintaining legitimacy within the staff group. Schedules needing frequent defence in corridor conversations were considered problematic, even if they formally complied with agreements. Predictability having recognisable weekly patterns was seen as equally important. Participants repeatedly stated that *"even fair schedules can feel unfair if they are constantly changing"*.

In practice, this meant that some local routines, such as "every fifth Friday is early for everyone in turn" or "certain people never open two days in a row", were treated as almost sacred, despite being undocumented. Several participants stressed that a system that ignored these unwritten norms would be perceived as unfair, regardless of how well it scored on quantitative fairness metrics.

5.5.3 Pedagogical Continuity and Child Stability

Staff continuity was critical for child wellbeing. Schedules that moved staff too frequently between groups were perceived as disruptive and pedagogically misaligned. Schedulers saw continuity as a constraint rather than an optimisation objective: certain constellations of teachers and child groups were considered non-negotiable anchors around which other duties had to be arranged.

Participants emphasised that children notice and react to instability. One preschool head described how a previous attempt to "spread staff evenly" across groups during a reorganisation had led to confusion and insecurity among younger children, who "kept looking for the grown-ups they were used to". This experience made some participants initially sceptical of any algorithmic approach that appeared to prioritise numeric fairness at the expense of continuity.

5.5.4 Trust and Transparency in Smart Systems

Transparency emerged as a prerequisite for trust. Participants needed to see:

- which rules were applied,
- when a rule was violated,
- why specific allocations were made.

Lack of such information led to suspicion and second-guessing even when schedules were objectively feasible. Several participants explicitly linked this to their accountability as managers: they needed to be able to explain and defend schedules not only to staff but also to unions, guardians, and municipal auditors. One administrator summarised this by saying: *"If I cannot see why the system did something, then I cannot stand behind it, even if it looks okay at first glance."*

5.5.5 Desire for Configurable Flexibility

Participants wanted the system to automate "routine reasoning" while allowing fine-grained adjustment of the final product. Configurable fairness rules, planning times, and break logic were seen as essential. Users expressed a preference for a small set of clearly understandable options rather than a large number of detailed sliders or hidden parameters.

In later iterations, this translated into a strong preference for strategies where the "shape" of the week remained recognisable but certain elements, such as Friday patterns or specific opening/closing duties, could be rotated with the help of the system. The overall expectation was that the tool should reduce manual effort without locking users into opaque, irreversible decisions.

6 Discussion

This study set out to iteratively design and evaluate a smart scheduling system for Swedish preschools, with a particular focus on how scheduling strategies and explainability features shape practitioners' perceptions of fairness, predictability, trust, and professional control. Across three design iterations involving fifteen participants, the findings consistently show that the central challenge is not the optimisation of constraints but the integration of algorithmic support into a sociotechnical environment defined by relational work, tacit norms, and pedagogical rhythms. This discussion therefore addresses the three research questions directly: how practitioners evaluate fairness and continuity in algorithmic outputs (RQ1), how explainability and rule feedback shape trust and accountability (RQ2), and how participatory prototyping enables efficiency gains while preserving professional judgement (RQ3). The analysis aligns with broader HCI and public-sector research suggesting that algorithmic systems operate as sociotechnical mediators embedded within institutional

practice, rather than purely technical optimisation tools [44, 45].

6.1 RQ1: Fairness, Predictability, and Pedagogical Continuity

The themes related to fairness, stability, organisational dynamics, and the challenge of defining a "good" schedule directly address RQ1. Together, they illustrate how practitioners' perceptions of workload distribution are inseparable from relational and pedagogical expectations.

Fairness as a Relational and Pedagogical Construct

Fairness was the dominant lens through which participants evaluated scheduling outcomes across all iterations. In relation to RQ1, the findings show that fairness was understood not only as an equitable distribution of duties but as a relational practice shaped by personal preferences, reciprocity, and pedagogical continuity. Although the fairness audit quantified workload distribution, participants emphasised that numerical balance alone does not constitute fairness. One preschool head noted that "even fair schedules can feel unfair if they are constantly changing," capturing how legitimacy depends on how schedules *feel* in everyday practice rather than how they score on abstract metrics. This aligns with organisational justice literature, which conceptualises fairness as socially negotiated and context dependent [46].

Participants also linked fairness to schedule predictability. Stable weekly patterns were perceived as inherently fair because they supported children's routines and reduced staff uncertainty. Local routines such as "every fifth Friday is early for everyone in turn" were treated as almost sacred, despite never being formally documented. These unwritten norms illustrate how fairness and predictability were evaluated together: a schedule that respected such patterns was considered more legitimate than one that redistributed duties perfectly on paper but disrupted local expectations. The preference for the hybrid strategy (Solution 3), where Fridays rotate while Monday–Thursday remain stable, illustrates that selective rotation rather than uniform fairness best aligned with practitioners' real-world expectations and existing informal practices.

The Difficulty of Defining a "Good" Schedule

The convergence of fairness, usability, and trust ratings suggests that practitioners experienced these criteria holistically: when a schedule felt good, it felt fair, predictable, and trustworthy at once. This finding relates to both RQ1 and RQ2, as it indicates that users do not evaluate the system along isolated dimensions. Instead, schedule quality emerges from the alignment of system logic with relational, pedagogical, and organisational expectations. In Iteration 3, some participants described Solution 2 as "fair mathematically, but not humane in practice," highlighting that schedules judged acceptable must balance numerical equity with everyday experiences of stability and workload.

This holistic evaluation also underscores that preferences are fluid and socially negotiated. During one Iteration 3 discussion, participants reconsidered which duties were most desirable. Late Mondays and early Fridays were initially framed as "luxury times" because they could extend the weekend. Another participant then noted that they preferred those shifts because they often involve fewer children and feel less demanding. Others immediately agreed, and time slots that had previously been treated as burdensome were reclassified as attractive. This illustrates that the perceived desirability of shifts is not fixed but emerges through collective interpretation, meaning that fairness cannot be derived from duty type alone. The study suggests that algorithmic scheduling systems must accommodate such evolving, context-dependent understandings of fairness rather than rely on static optimisation objectives.

Epistemic Boundaries in Evaluating Algorithmic Schedules

A methodological limitation relevant to RQ1 and RQ3 is that most participants were schedulers or preschool heads rather than the staff who experience schedules directly in day-to-day work. Their assessments were therefore second-hand and filtered through managerial interpretation. Several leaders explicitly noted that much of the informal feedback they receive from staff comes in the form of complaints rather than balanced reflections, making it difficult to distinguish isolated frustrations from more systemic issues. As one head remarked during Iteration 1, "you mostly hear about the days that went badly, not the weeks that worked fine," highlighting how managerial perspectives can become shaped by a narrow and emotionally charged subset of experiences. This may influence how fairness, predictability, and efficiency were evaluated in this study: leaders often design around the complaints they

hear, while quieter forms of satisfaction or stability remain underrepresented.

Another epistemic gap concerns the absence of municipal administrators, the actors ultimately responsible for procurement of such a system. Their priorities may differ substantially from those of preschool-level practitioners. In the context of nationwide downsizing [47, 48], municipal stakeholders increasingly emphasise cost-efficiency, auditability, and measurable outputs. Several participants alluded to this dynamic indirectly, noting that local decisions often need to be justified "upwards" and that tools capable of producing traceable, standardised reasoning would likely appeal to municipal supervisors. Such institutional incentives may lead municipalities to favour algorithmic solutions that prioritise efficiency metrics or uniformity across units, potentially creating tensions with the relational, context-sensitive fairness norms created by preschool leaders. Future research should therefore include frontline staff as well as municipal decision-makers to understand how algorithmic scheduling is negotiated across institutional layers with differing realities and responsibilities.

Organisational Dynamics, and the Politics of Scheduling

Scheduling is inherently political: it allocates workload, pedagogical responsibility, breaks, planning time, and access to desirable shifts. Participants repeatedly described scheduling as "a puzzle that only works once you've done it for five years," pointing to how expertise accumulates through tacit and relational knowledge rather than formal rules. The introduction of algorithmic outputs made several of these politics newly visible. By surfacing hidden patterns such as uneven distribution of opening or closing shifts, or habitual assumptions about which staff members could "handle" certain duties—the system challenged long-standing practices that had previously gone unquestioned. In one workshop, a participant reacted with surprise when the fairness audit revealed that a staff member had been closing far more often than anyone realised, saying: "We knew it was unbalanced, but not *that* unbalanced." Such moments illustrate how algorithmic visibility can disrupt local norms, empower practitioners to question inherited routines, and introduce new forms of accountability.

However, this increased visibility also raised concerns about overstandardisation. When practices become datafied, informal and relational nuances risk

being overshadowed by metrics that privilege consistency or efficiency. Several participants worried that a schedule that "looks fair on paper" might still undermine pedagogical needs, such as maintaining stable teaching teams or minimising transitions for younger children. As one scheduler explained, "even fair schedules can feel unfair if they move people around too much." These concerns connect to broader issues of trust: visibility can legitimise decisions, but it can also generate pressure to justify choices according to measurable criteria rather than professional judgement.

Participants consistently emphasised that pedagogical care rhythms, such as children's need for continuity with familiar adults must remain central. These rhythms do not align neatly with optimisation logic, and several leaders recounted past experiences where attempts to "spread staff evenly" had inadvertently created anxiety among children. This tension shows that algorithmic scheduling must continually negotiate between organisational performance goals and the relational foundations of early childhood education. Moreover, the risk of "locking in" outdated or context-specific constraints was a recurring theme. Without mechanisms for revisiting and adjusting constraints, an algorithmic system may inadvertently perpetuate compromises or inequities from earlier periods. Such risks highlight the need for ongoing governance and participatory recalibration. Maintaining fairness and relevance over time requires not only technical robustness but organisational stewardship, an emergent role that many participants felt their teams were not yet formally prepared to assume.

6.2 RQ2: Explainability, Trust, and Accountability

Themes addressing explainability, governance, and system legibility speak directly to RQ2. They show that trust depends not on accuracy but on visibility, interpretability, and the ability to justify decisions within organisational structures.

Trust and Legibility in Human–System Collaboration

In relation to RQ2, trust was strongly tied to system legibility. Participants trusted the system when they could see how constraints interacted, understand why specific assignments were made, or inspect rule violations. Rather than expecting the system to be "correct," participants expected it to be explain-

able. One administrator put this succinctly: "If the system tells me exactly what is wrong and why, then I feel I can stand behind the schedule." Another echoed a similar sentiment: "If I cannot see why the system did something, then I cannot stand behind it, even if it looks okay at first glance." These comments reflect broader insights from explainable AI, where trust is relational and grounded in interpretability rather than accuracy [26, 27, 20].

Explainability also served an organisational function. Practitioners needed to justify schedules to colleagues, unions, guardians, and municipal actors. Visible rule feedback and warning panels were described as making it easier to "stand for" a schedule, because they externalised the reasoning that had previously been held in the heads of experienced schedulers. The repeated description of the tool as a "thinking partner" reinforces that practitioners did not expect automation to replace judgement but to enhance reflective decision-making through visible reasoning. In this sense, interpretability was not an optional feature but a prerequisite for delegating any meaningful part of the scheduling task to the system.

Governance, Algorithmic Caretaking, and Institutional Responsibility

Introducing algorithmic scheduling raises governance and accountability questions relevant to both RQ2 and RQ3. Participants emphasised the need for reproducible, auditable behaviour, aligning with EU AI Act expectations for high-risk systems. Trust was reinforced by deterministic outputs—"the same inputs always give the same outputs"—because this predictability supports accountability and reduces cognitive workload for schedulers who must justify their choices. At the same time, the results showed that constraint configurations themselves quickly became a site of negotiation: Iteration 3 workshops often began with discussions about "which fairness rules we actually stand for," indicating that governance is as much about value setting as it is about technical tuning.

Participants also stressed that someone must maintain constraints, interpret warnings, and mediate system behaviour, an emergent role aligning with literature on algorithmic stewardship [45]. Yet several leaders admitted that this stewardship role is not formally recognised in current organisational structures, even though they already spend evenings "just moving boxes around"

in spreadsheets. Autonomy must be preserved; if constraints are locked or centralised, practitioners lose the ability to negotiate fairness and adapt to local practices. These findings reinforce the need for configurability, clear responsibility for constraint maintenance, and institutional recognition of the interpretive labour involved in caring for algorithmic systems.

6.3 RQ3: Tacit Knowledge, Prototyping, and Professional Judgement

Themes addressing tacit knowledge, alternative proposals, interpretive labour, and system configurability align with RQ3. They show how participatory prototyping supports efficiency gains while preserving professional control.

Tacit Knowledge and the Interpretive Labour of Scheduling

Tacit knowledge played a central role in shaping how participants interpreted both system outputs and design iterations. This theme connects directly to RQ3: iterative participatory prototyping was effective not because users specified perfect rules, but because it revealed the tacit, experiential knowledge that needed to be encoded or deliberately left flexible. Participants repeatedly described scheduling as "a puzzle that only works once you've done it for five years," emphasising that much competence emerges only after years of relational, embodied practice.

The Results chapter showed how this tacit expertise manifests in everyday work: preschool heads spending evenings "just moving boxes around" in spreadsheets, relying on colour codes and shorthand that only they understand, and feeling personally responsible for keeping peace in the staff group. Prototyping helped surface this invisible labour by making assumptions explicit and inviting discussion about which practices should be preserved, formalised, or reconsidered. Several participants suggested that the system could serve as a form of institutional memory. A way of documenting practices that would otherwise "leave with the person" when an experienced scheduler changes workplace. This supports the conclusion that human–system collaboration must preserve professional judgement and interpretive flexibility, rather than attempt to formalise all context-specific practices into fixed rules.

The Value of Rapid Exploration and Multiple Algorithmic Proposals

The ability to generate multiple schedule proposals quickly was highly valued and directly addresses RQ3. Rather than functioning as a final decision-maker, the system supported comparative reasoning and collaborative negotiation. Participants described the automatically generated rosters as "a good skeleton, but still in need of flesh," indicating that they saw outputs as robust starting points rather than finished products. The side-by-side comparison of the three multi-week strategies in Iteration 3 made trade-offs immediately visible: participants could see at a glance who opened, who closed, how "nice Fridays" were distributed, and how often pedagogical anchors were disturbed.

Participants viewed these alternatives as tools for reducing interpersonal tension and making trade-offs explicit—important for maintaining professional judgement while reducing administrative workload. One of the key benefits was that negotiation could shift from defending individual decisions ("why did I get this Friday?") to discussing properties of different proposals ("in this version, Fridays are fair but the weeks feel messy"). This supports the idea that presenting alternatives can strengthen collective sensemaking and perceived legitimacy by making trade-offs discussable rather than implicit. Rapid generation therefore contributes both to efficiency and to organisational transparency, by turning scheduling into a structured exploration of options rather than a fragile, undocumented compromise.

6.4 Algorithmic Scheduling as Sociotechnical Care Infrastructure

Across all data sources, the study shows that algorithmic scheduling systems become part of a broader care infrastructure integrating relational negotiation, pedagogical continuity, institutional governance, and computational structure. Viewed through the lens of the research questions:

- Fairness and predictability (RQ1) depend on stability, relational norms, and alignment with children's rhythms.
- Trust and accountability (RQ2) depend on legibility, explainability, and the ability to justify decisions to stakeholders.

- Reduced administrative burden (RQ3) emerges only when the system enhances and not replaces tacit expertise and professional agency.

Together, these insights can conceptualise algorithmic scheduling not as a means of optimisation but as a sociotechnical mediator that must balance clarity with flexibility and structure with context.

6.5 Notes to Stakeholders

The findings of this study offer several implications for Tempus as it continues developing algorithmic scheduling tools for preschools. Across all three design iterations, practitioners expressed a strong preference for systems that respect the relational and pedagogical contexts in which scheduling occurs. Rather than functioning as optimisation engines, scheduling tools were perceived as most valuable when they operated as collaborative companions clarifying reasoning, supporting negotiation, and preserving professional judgement. These expectations suggest that future development should emphasise explainability, configurability, and flexibility as foundational product features, not optional enhancements.

A key insight is the importance of maintaining weekly stability while enabling long-term fairness. Practitioners rejected approaches that rebalanced duties aggressively from week to week, even when mathematically fair, because they disrupted children's routines and teachers' planning rhythms. The hybrid strategy emerged as the only model perceived as both fair and operationally realistic. For Tempus, this underscores the value of designing scheduling algorithms that privilege continuity of pedagogical care as a primary constraint, and that treat fairness as a temporal process unfolding over weeks rather than as a static numerical target.

Equally important is the need for transparency in constraint behaviour. Participants repeatedly highlighted that they must be able to understand why a particular assignment was made, especially when schedules are shared with colleagues or parents. Explainability is therefore not a cosmetic feature but a core requirement for trust, adoption, and troubleshooting. This may involve interface elements such as traceable decision paths ("this shift was assigned because..."), visual explanations of rule interactions or violations, and sensitivity indicators showing how small changes in constraints affect outcomes.

For Tempus, investing in explainability features not only improves usability but reduces support burden by externalising decision logic directly to users.

The study also highlights the importance of embedding flexibility and local autonomy into system configuration. Practitioners operate with nuanced combinations of tacit norms, interpersonal agreements, and pedagogical priorities that cannot be fully captured in predefined templates. Tools that restrict or centralise control over constraints risk undermining the core social processes of scheduling. By contrast, systems that allow administrators to tune weights, toggle constraints, or create local exceptions empower teams to maintain fairness while adapting to context-specific needs. From a technical perspective, this suggests the value of modular constraint architectures with clear metadata, versioning, and safeguarded editing interfaces.

Another strong practitioner preference was for editable drafts and multiple algorithmic proposals. Tempus should consider workflows where the system is not framed as producing "the schedule" but as generating candidate solutions for human comparison. This aligns well with optimisation strategies that can produce diverse near-optimal solutions quickly, and with UX patterns that help users reason across alternatives. It also reduces conflict in scheduling discussions by relocating negotiation from personal preference to a shared artefact: the set of proposals.

Finally, the study suggests opportunities for Tempus to support tacit knowledge rather than attempting to formalise it away. Participants consistently described the system as valuable when it surfaced patterns, provided memory support, or made hidden dynamics visible. Participants valued the system when it surfaced patterns and supported memory, but resisted attempts to rigidify tacit practices into fixed rules. Instead, they preferred that local norms could be expressed as adjustable preferences, guiding the system without fully constraining it. A successful product strategy could therefore treat tacit knowledge as an asset to be surfaced and scaffolded through different tools such as contextual warnings, role-specific customisation, and features that allow users to embed local knowledge directly into the system.

Taken together, these findings suggest that Tempus has an opportunity to position its scheduling system not simply as an optimisation engine but as a

sociotechnical facilitator one that supports relational fairness, organisational transparency, and contextual professional judgement. By designing for legibility, configurability, and collaborative control, Tempus can build a system that aligns with practitioners' needs while maintaining the technical robustness required for large-scale deployment.

7 Conclusion

This thesis investigated how algorithmic decision support can be meaningfully integrated into the everyday scheduling practices of Swedish preschools. Through three iterative cycles of design, prototyping, and evaluation, the study examined how legal, organisational, pedagogical, and cultural constraints can be formalised within a computational model, and how such a model can be made transparent, configurable, and trustworthy for its intended users. By grounding design decisions in participatory evaluation, the project explored not only what a schedule optimiser can compute, but what practitioners consider acceptable, fair, and pedagogically sound.

Across all phases, the findings show that successful algorithmic scheduling in preschools depends far more on alignment with professional norms and user expectations than on optimisation performance alone. Preschool scheduling is a deeply situated activity shaped by pedagogical rhythms, fairness norms, tacit knowledge, and organisational responsibility. The smart scheduling prototype developed in this project demonstrates that algorithmic reasoning can support this work, but only when embedded in an interface that clearly communicates constraint logic, enables human oversight, and respects established working cultures. Rather than merely automating scheduling, the system must make its reasoning visible, reduce repetitive work, and provide alternative scenarios for deliberation.

The study produced three main contributions. First, it translated legal rules, staffing requirements, pedagogical needs, and local operational constraints into a transparent and interpretable constraint model. The development of hard and soft constraints, together with configurable weights, showed that computational formalisation is feasible without sacrificing clarity. This work contributes to public-sector algorithm design by demonstrating that constraint-based approaches can encode complex sociotechnical requirements while remaining explainable to non-technical users.

Second, the study demonstrated the centrality of explainability for trust and acceptance. Participants consistently emphasised the importance of visible rule feedback, warnings, and rationales for system decisions. These mechanisms supported both accountability toward staff, unions, and municipal authorities and users' cognitive understanding of how schedules were produced. Clear

explanations were perceived as essential even when schedules were judged realistic and fair, highlighting that transparency is not a secondary feature but a prerequisite for the legitimacy of algorithmic decision support in pedagogical contexts.

Third, the evaluation of three alternative multi-week scheduling strategies revealed how different algorithmic priorities shape practitioners' perceptions of fairness, realism, and applicability. While coverage-oriented strategies offered stability and rotation-oriented strategies offered equity, participants preferred hybrid approaches that preserved weekly recognisability while distributing less desirable tasks proportionally over time. These findings underscore that fairness in preschool scheduling is not purely quantitative; it is deeply cultural, negotiated, and relational. Fairness must therefore be considered as an evolving constellation of expectations rather than a static optimisation target.

Taken together, the results illustrate that smart scheduling in preschools is best understood as a sociotechnical practice: a collaboration between human judgement and computational reasoning. Algorithmic tools can reduce administrative workload, increase transparency, and support more defensible decision-making provided that users retain meaningful control and that system behaviour is understandable, configurable, and aligned with pedagogical values. The study contributes to HCI and public-sector AI research by showing that the design of algorithmic tools must account for tacit expertise, interpretive labour, and the politics of work distribution.

The work also highlights that algorithmic design for public-sector contexts cannot be separated from considerations of law, ethics, accessibility, and professional culture. Compliance requirements such as traceability and human oversight, as emphasised in the EU AI Act, can be met not only through documentation but through interaction design choices that foreground explainability, configurability, and stewardship. This reframes compliance as a design challenge rather than solely a documentation requirement.

Overall, this thesis demonstrates that smart scheduling has the potential to strengthen organisational sustainability in Swedish preschools by supporting fairness, reducing administrative burden, and making scheduling decisions more transparent. However, its successful adoption depends on careful atten-

tion to local practices, relational norms, and pedagogical aims. Algorithmic support must therefore be seen not only as a technical optimisation problem but as an ongoing process of negotiation between computational models and the lived realities of preschool work.

8 Future Work

The findings of this study point to several promising directions for further research and development. A central insight concerns the role of constraints both formal and informal in shaping the quality, fairness, and acceptability of schedules. Participants repeatedly expressed the desire to encode additional rules, toggle alternative interpretations of fairness, and explore the consequences of different configurations. These needs suggest that future systems should move beyond fixed constraint sets toward flexible, extensible, and user-driven constraint architectures that foreground configurability as a core design principle rather than an advanced feature.

Toward configurable and extensible constraint systems. A key direction for future work is the design of interfaces and computational models that allow practitioners to create, modify, and combine constraints without technical expertise. Instead of hard-coding scheduling logic, systems could provide libraries of reusable constraint modules covering pedagogical rules, labour regulations, fairness logics, and local norms—that users can activate or adapt. Such modularity would respect the diversity of preschool practices across municipalities while maintaining a coherent computational foundation.

Exploring design spaces through systematic variation. Participants also expressed a desire to compare alternative strategies quickly and transparently. Building on this, future systems could incorporate design-space exploration tools that allow users to filter, generate, and evaluate multiple schedules simultaneously. Users might toggle between alternatives that prioritise continuity, fairness, or staff preferences, examining differences through visualisations of coverage, stability, or pedagogical patterns.

Support for soft constraints and relational knowledge. Another direction concerns the representation of soft constraints such as tacit norms, interpersonal dynamics, and relational pedagogical considerations. Future work could investigate hybrid approaches that combine explicit constraint satisfaction with structured annotations, pattern-based heuristics, or rule-of-thumb templates co-created with practitioners. These might include rules such as "avoid pairing inexperienced staff alone during open," "favour familiar adults during transitions," or "balance relational load for staff supporting children with additional needs."

Such relational knowledge is difficult to formalise but central to pedagogical quality. Further research is needed to determine how to encode soft constraints without introducing unintended bias or diminishing practitioner autonomy.

Learning from use: adaptive but controllable systems. Although this study avoided machine-learning approaches due to governance and accountability concerns, future work may explore controlled forms of adaptivity that remain transparent and user-controlled. For example, systems could learn which constraints are frequently overridden, where conflicts commonly arise, or which configurations are consistently preferred. These insights could inform meta-design features such as suggesting constraint bundles, highlighting fragile parts of schedules, or generating alternative scenarios tailored to local practice.

Any adaptive functionality must, however, remain auditable and deterministic, ensuring compliance with public-sector requirements and avoiding the opacity associated with high-risk AI systems.

Integrating forecasting and real-time data. Another opportunity lies in integrating scheduling tools with attendance forecasting or real-time municipal data streams. Predictive information about child attendance, staff availability, or seasonal variation could help teams anticipate fragility and prepare for contingencies. Systems might generate stability indicators or scenario warnings that complement rather than undermine weekly continuity.

The challenge will be to introduce responsiveness without compromising the predictability that practitioners value.

Long-term deployment studies and organisational transformation. Finally, future work should investigate how algorithmic scheduling tools function in long-term, real-world deployment. Questions remain about how norms evolve through continued system use, how constraint sets drift or stabilise, how new staff learn and appropriate the system, and how municipalities govern updates over time. Multi-site case studies would provide important insights into organisational adoption, path dependency, and the sustainability of algorithmic care infrastructures.

Together, these directions point toward a broader research agenda that situates algorithmic scheduling within the ecosystems of labour relations, pedagogical practice, and public-sector digitalisation. By enabling flexible constraint creation, supporting design-space exploration, and embedding systems in real workflows, future work can deepen both the technical and sociotechnical foundations of algorithmic support in early childhood education.

Summary of Contributions

This thesis makes three primary contributions to research on algorithmic decision support in public-sector and educational contexts:

1. A transparent and interpretable constraint model for preschool scheduling.
The thesis translates legal regulations, pedagogical needs, staffing requirements, and local norms into a structured constraint model that remains understandable to non-technical users. This demonstrates that sophisticated scheduling logic can be formalised without sacrificing explainability.

2. Empirical evidence on the role of explainability, trust, and configurability in algorithmic scheduling. Through participatory prototyping with practitioners, the project shows that trust depends less on accuracy than on visible reasoning, rule feedback, and opportunities for meaningful oversight. The study advances knowledge of how explainability supports accountability and professional confidence in public-sector algorithmic tools.

3. Design insights into fairness, continuity, and hybrid scheduling strategies.
By evaluating three scheduling strategies, the study reveals how practitioners negotiate fairness and predictability and why hybrid approaches balancing weekly stability and long-term equity are preferred over purely optimisation-driven solutions. These findings position algorithmic scheduling as a sociotechnical practice requiring ongoing negotiation between computational reasoning and pedagogical judgement.

Together, these contributions advance both the technical design and the sociotechnical understanding of algorithmic support for early childhood education.

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Appendices

A Interview Guide – Iteration 1

The following questions were used during the first round of semi-structured interviews with preschool administrators. The purpose of this interview was to understand current scheduling practices, challenges, and expectations regarding digital decision-support tools.

Questions

1. Can you describe how you currently create and manage staff schedules at your preschool?
2. What are the main challenges you experience with your current scheduling process?
3. How do you balance pedagogical, legal, and fairness considerations when making schedules?
4. In what ways do digital tools (like Tempus Admin) help or limit you today?
5. How do you feel about the idea of a system that automatically suggests or generates schedules?
6. What kind of control or transparency would you need to feel comfortable using such a system?
7. Is there anything else you would like to add about how scheduling could be improved?

B Interview Guide – Iteration 2

The second interview guide focused on evaluating early prototypes of the smart scheduling function and gathering feedback on usability, trust, and perceived usefulness.

Questions

1. How did you experience interacting with the prototype compared to your current scheduling tools?
2. Were the system's suggestions clear and understandable? Why or why not?
3. Did the explanations provided by the system help you trust its suggestions?
4. Which parts of the interface felt most intuitive, and which were confusing or unnecessary?
5. How well do you think the system supports your daily work tasks?
6. What would you like to see improved or added in future versions?
7. Overall, would you consider using a system like this in your preschool? Why or why not?

Interpretation: Together, these figures demonstrate the visual and interactive evolution of the prototype across the three design cycles. They also illustrate how iterative feedback from participants progressively shaped system transparency, configurability, and fairness mechanisms. Each visual artefact functioned as a conversational prompt during workshops, enabling users to externalise expectations and evaluate the system's reasoning processes.

C Interview Guide – Iteration 3

The third interview guide focused on the comparative evaluation of refined prototype versions and explored participants' perceptions of fairness, usability, and explainability. This iteration combined semi-structured discussion with short post-session Likert-scale questionnaires to collect both qualitative and quantitative feedback.

Semi-Structured Interview Questions

1. When comparing the different automatically generated schedules, which version felt most realistic or useful for your daily work? Why?
2. How would you describe the balance between fairness and practicality in the generated rosters?
3. Did the explanations provided by the system help you understand how scheduling decisions were made?
4. How confident would you feel using this system to produce schedules independently?
5. Were there any moments where the system's logic felt unclear or contradictory to your expectations?
6. How could the interface or explanation features be improved to better support your decision-making?
7. Is there anything that would make you more likely to adopt or trust this type of system in your preschool context?

Post-Session Likert-Scale Questionnaire

After each interview, participants completed a short Likert-scale form rating their agreement with several statements on a five-point scale (*1 = Strongly disagree ... 5 = Strongly agree*). The aim was to quantify perceived fairness, usability, and transparency of the generated schedules.

Table 3 – Illustrative example of the Likert-scale questionnaire layout used in Iteration 3. The actual form was provided digitally to participants and mirrored this visual structure.

Statement	1	2	3	4	5
The generated schedule felt fair in terms of shift distribution.	☐	☐	☐	☐	☐
I would want to use this system to assist in real preschool scheduling..	☐	☐	☐	☐	☐
I would trust this system to assist in real preschool scheduling.	☐	☐	☐	☐	☐